

Fixed-multiplicity shuffles and balanced Smirnov words

The $e^{-(k-1)}$ law, Poisson adjacency statistics,
and explicit all-orders corrections

A research note on OEIS A007060, A193624, A330266, and related sequences

1 October 2026

Abstract

Let $B_{n,k}$ be the number of permutations of kn distinct cards, partitioned into n ranks of size k , in which neighboring cards never have the same rank. The OEIS entry A330266 [1] records the fixed- k conjecture (S. Kirgizov, 29 September 2023)

$$\frac{B_{n,k}}{(kn)!} \longrightarrow e^{-(k-1)}.$$

The case $k = 4$ was subsequently deduced on that entry from an approximation to a related sequence (D. Radcliffe, 9 September 2025), but the general statement remained posed there. We give a direct proof for every fixed $k \geq 2$ and prove considerably more. If $X_{n,k}$ is the number of equal-rank adjacencies in a uniformly random shuffle, then

$$X_{n,k} \xrightarrow{d} \text{Poisson}(k-1).$$

We derive an exact probability-generating integral whose kernel is an associated Laguerre polynomial, an exact inclusion–exclusion formula, a gamma-expectation representation, and a constructive expansion to arbitrary order. In particular, uniformly for fixed k ,

$$\begin{aligned} \frac{B_{n,k}}{(kn)!} = e^{-(k-1)} & \left[1 - \frac{(k-1)^2}{2kn} + \frac{(k-1)^2(3k^2 - 14k + 7)}{24k^2n^2} \right. \\ & \left. - \frac{(k-1)^4(k^2 - 10k + 17)}{48k^3n^3} + O_k(n^{-4}) \right]. \end{aligned}$$

The first four factorial cumulants are obtained in exact rational form; the mean is exactly $k-1$ and the variance is $(k-1) - (k-1)^2/(kn-1)$. For A330266 this gives

$$\frac{B_{n,4}}{(4n)!} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}) \right).$$

We also treat the corresponding multiset sequences, local probabilities, Hamiltonian paths in complete multipartite graphs, ratio asymptotics, and a Lambert- W inverse transseries. Exact reproducibility code accompanies the article.

Scope and priority. The sequence definitions and historical comments below are taken from the OEIS pages cited in the bibliography. A330266 [1] now states that its particular $k = 4$ leading equivalent follows from an approximation to A190830. The point of the present note is the uniform fixed- k theorem, the Poisson law, exact factorial cumulants, and systematic correction terms. No claim of a complete literature search or historical priority is made; the results should be independently checked before submission or attribution.

Editorial note (1 October 2026, batch 73O2). Manuscript 57 of batch 73 of ProveIt’s incoming-reports intake, placed in commit `6e193dd4f`; AI-assisted, unrefereed, not formalized. The write added the label prefix `bsw:`, citation commands, the names of the OEIS contributors, notes marked “[Added 1 October 2026, batch 73O2]”, Remark 6.3, one bibliography entry and Appendix C. The analytic tail step of Theorem 6.2 is a sketch; the fixed- k limit law does not depend on it. [2 October 2026, batch 77P5.] Part II, from manuscript 11 of batch 77, proves that step for fixed k (Remark II.6.3); Sections 1–15 and Appendices A–C form Part I. [5 October 2026, batch 101.] Part III, from bundle Report 98, proves the same step a second time, written from the same pin without knowledge of Part II, and adds the n^{-4} terms (Section III.7).

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1 The OEIS problem and the family of sequences

Fix integers $n \geq 0$ and $k \geq 2$. There are n ranks, each containing k distinct cards. A shuffle is a permutation of all $N = kn$ cards. Let

$$X_{n,k} = \#\{1 \leq j < N : \text{the cards at positions } j \text{ and } j+1 \text{ have the same rank}\}. \quad (1)$$

Write

$$B_{n,k} = N! \mathbb{P}(X_{n,k} = 0), \quad p_{n,k} = \frac{B_{n,k}}{N!}. \quad (2)$$

If cards within one rank are made indistinguishable, let $S_{n,k}$ denote the number of admissible multiset words. Every rank word lifts to exactly $(k!)^n$ labeled permutations, so

$$B_{n,k} = (k!)^n S_{n,k}, \quad p_{n,k} = \frac{S_{n,k}}{(kn)!/(k!)^n}. \quad (3)$$

Thus the same probability governs the labeled and multiset versions.

The most visible OEIS rows are as follows.

k	labeled count $B_{n,k}$	interpretation	multiset count $S_{n,k}$
2	A007060 [2]	couples with no adjacent spouses	A114938 [4]
3	A193624 [3]	triples with no adjacent siblings	A193638 [5]
4	A330266 [1]	four cards in each rank	A321633 [6]

A330266 [1] records the exact integral

$$B_{n,4} = \int_0^\infty (t^4 - 12t^3 + 36t^2 - 24t)^n e^{-t} dt$$

and the conjectured fixed-multiplicity law, posed there by S. Kirgizov on 29 September 2023,

$$p_{n,k} \longrightarrow e^{-(k-1)}. \quad (4)$$

The entry also records a September 2025 deduction of the $k = 4$ equivalent from a separate approximation: D. Radcliffe (9 September 2025) combined A330266(n) = $24^n n!$ A190830(n) with V. Kotěšovec's approximation of A190830 and Stirling's formula. Our derivation below is direct, applies to every fixed k , and identifies the full limiting distribution behind (4).

1.1 Why a Poisson law is plausible but not automatic

For each of the $N - 1$ adjacent position pairs, the probability of equal ranks is $(k - 1)/(N - 1)$. Hence

$$\mathbb{E}X_{n,k} = (N - 1) \frac{k - 1}{N - 1} = k - 1. \quad (5)$$

If the rare adjacency indicators were independent, one would expect a Poisson limit of mean $k - 1$ and hence a zero probability $e^{-(k-1)}$. They are not independent: overlapping position pairs and the fixed rank multiplicities produce both local and global dependence. The proof therefore requires more than the heuristic “rare events with bounded mean.” The exact generating function will make the cancellation transparent.

1.2 Hamiltonian-path interpretation

Replace the deck by the complete n -partite graph with n parts of size k . A vertex ordering is an admissible shuffle precisely when consecutive vertices are joined. Consequently $B_{n,k}$ is the number of *oriented* Hamiltonian paths, counted as ordered vertex lists, in $K_{k,k,\dots,k}$. Unoriented paths are counted by $B_{n,k}/2$ when $kn > 1$. This connection places the problem among balanced Smirnov words and complete multipartite Hamiltonian enumeration [7, 9].

2 A weighted run transform

It is useful to enumerate all shuffles while recording the number of bad adjacencies. Define

$$A_{n,k}(u) = \sum_{\pi \in \mathfrak{S}_{kn}} u^{X_{n,k}(\pi)}, \quad G_{n,k}(u) = \frac{A_{n,k}(u)}{(kn)!} = \mathbb{E} u^{X_{n,k}}. \quad (6)$$

Thus $A_{n,k}(0) = B_{n,k}$ and $G_{n,k}(0) = p_{n,k}$.

Theorem 2.1 (Run-transform generating function). *For indeterminates $x_1, \dots, x_n$,*

$$\sum_w u^{X(w)} x_1^{|w|_1} \dots x_n^{|w|_n} = \frac{1}{1 - \sum_{i=1}^n \frac{x_i}{1 + (1-u)x_i}}, \quad (7)$$

where the sum is over all finite words on $\{1, \dots, n\}$, including the empty word. Therefore

$$A_{n,k}(u) = (k!)^n [x_1^k \dots x_n^k] \frac{1}{1 - \sum_{i=1}^n \frac{x_i}{1 + (1-u)x_i}}. \quad (8)$$

Proof. Compress every maximal run of equal letters to one letter. The compressed word is a Smirnov word, meaning that consecutive letters are distinct. A run of letter i and length $r \geq 1$ contributes $u^{r-1} x_i^r$, whose generating series is

$$y_i = \sum_{r \geq 1} u^{r-1} x_i^r = \frac{x_i}{1 - u x_i}.$$

The classical multivariate generating function of Smirnov words [12, 13] is

$$\frac{1}{1 - \sum_i y_i / (1 + y_i)}.$$

Indeed, it follows from the ordinary word series $1/(1 - \sum_i z_i)$ by replacing each nonempty run by one marked letter, or by the standard run-decomposition bijection. Substitution gives

$$\frac{y_i}{1 + y_i} = \frac{x_i}{1 + (1-u)x_i},$$

which proves (7). Extracting k occurrences of each rank gives multiset words; labeling the k copies independently in each rank contributes $(k!)^n$ and proves (8). $\square$

3 The exact Laguerre integral

Set $v = u - 1$ and define

$$P_{k,u}(t) = k! [x^k] \exp\left(\frac{tx}{1-vx}\right). \quad (9)$$

Theorem 3.1 (Exact integral and polynomial kernel). *For every $n \geq 0$, $k \geq 2$, and complex u ,*

$$A_{n,k}(u) = \int_0^\infty e^{-t} P_{k,u}(t)^n dt. \quad (10)$$

The kernel has the explicit forms

$$P_{k,u}(t) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{k!}{j!} t^j (u-1)^{k-j}, \quad (11)$$

$$= k!(u-1)^k L_k^{(-1)}\left(-\frac{t}{u-1}\right), \quad (12)$$

where the value at $u = 1$ is interpreted by continuity.

Proof. Use the coefficientwise Laplace identity

$$\frac{1}{1-s} = \int_0^\infty e^{-t} e^{ts} dt.$$

In (8), the exponential factors into n identical univariate series. Extracting x_i^k from each factor gives (10). Expanding $\exp(tx/(1-vx))$ first by the exponential degree and then extracting x^k gives (11). Finally, the standard expansion [14]

$$L_k^{(-1)}(z) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{(-z)^j}{j!}$$

gives (12). □

For no equal adjacencies, $u = 0$ and $v = -1$. Write

$$P_k(t) := P_{k,0}(t) = (-1)^k k! L_k^{(-1)}(t). \quad (13)$$

The first kernels are

$$\begin{aligned} P_2(t) &= t^2 - 2t, \\ P_3(t) &= t^3 - 6t^2 + 6t, \\ P_4(t) &= t^4 - 12t^3 + 36t^2 - 24t, \\ P_5(t) &= t^5 - 20t^4 + 120t^3 - 240t^2 + 120t. \end{aligned}$$

The third line recovers the integral printed on A330266, while the theorem explains it as one row of a uniform Laguerre family.

4 Exact inclusion–exclusion and a compact reciprocal polynomial

For $0 \leq m \leq k-1$, put

$$a_{k,m} = \binom{k-1}{m} \frac{k!}{(k-m)!} = m! \binom{k}{m} \binom{k-1}{m}, \quad (14)$$

and define the degree- $(k-1)$ polynomial

$$R_k(z) = \sum_{m=0}^{k-1} a_{k,m} z^m. \quad (15)$$

The no-adjacency reciprocal polynomial is

$$C_k(z) = z^k P_k(1/z) = R_k(-z). \quad (16)$$

Theorem 4.1 (Exact finite formula). *Let $N = kn$. Then*

$$B_{n,k} = \sum_{m_1, \dots, m_n=0}^{k-1} (-1)^M (N-M)! \prod_{i=1}^n a_{k,m_i}, \quad M = m_1 + \dots + m_n, \quad (17)$$

$$= \sum_{M=0}^{n(k-1)} (N-M)! [z^M] C_k(z)^n. \quad (18)$$

More generally, the factorial probability generating function

$$\Phi_{n,k}(v) := \mathbb{E}(1+v)^{X_{n,k}} = G_{n,k}(1+v) \quad (19)$$

satisfies the exact identity

$$\Phi_{n,k}(v) = \sum_{M=0}^{n(k-1)} \frac{[z^M] R_k(z)^n}{(N)_{\underline{M}}} v^M. \quad (20)$$

Proof. From (11),

$$z^k P_{k,u}(1/z) = \sum_{m=0}^{k-1} a_{k,m} v^m z^m = R_k(vz).$$

Raise to the n th power and integrate termwise in (10). Since $\int_0^\infty t^{N-M} e^{-t} dt = (N-M)!$, division by $N!$ yields (20). Setting $v = -1$ gives (17) and (18). $\square$

Formula (18) is a rook-polynomial form of inclusion–exclusion. It uses only a degree- $(k-1)$ polynomial raised to the n th power and is efficient for exact integer computation. In particular, for $k = 4$ it reproduces

$$B_{13,4} = 3668033946384704437729512814619767610579526911188666362431432294400,$$

the standard-deck value recorded on A330266.

5 The Poisson adjacency theorem

Recall that the r th falling factorial is $(x)_{\underline{r}} = x(x-1)\cdots(x-r+1)$.

Theorem 5.1 (Factorial moments). *For fixed $k \geq 2$ and fixed $r \geq 0$,*

$$\mathbb{E}(X_{n,k})_{\underline{r}} = (k-1)^r + O_{k,r}(n^{-1}). \quad (21)$$

Consequently,

$$\boxed{X_{n,k} \xrightarrow[n \rightarrow \infty]{d} \text{Poisson}(k-1).} \quad (22)$$

Proof. The factorial moment is $r!$ times the coefficient of v^r in (20). Since every term $a_{k,m}z^m$ in $R_k(z)$ has the same z -degree as the corresponding v -degree, the denominator for this coefficient is exactly $(N)_{\underline{r}}$. The leading contribution to $[z^r]R_k(z)^n$ selects $a_{k,1}z$ from r distinct factors and 1 from the rest:

$$[z^r]R_k(z)^n = \binom{n}{r} a_{k,1}^r + O_{k,r}(n^{r-1}).$$

Here $a_{k,1} = k(k-1)$. Therefore

$$\begin{aligned} \mathbb{E}(X_{n,k})_{\underline{r}} &= \frac{r! \binom{n}{r} [k(k-1)]^r + O_{k,r}(n^{r-1})}{(kn)_{\underline{r}}} \\ &= (k-1)^r + O_{k,r}(n^{-1}). \end{aligned}$$

These are the factorial moments of $\text{Poisson}(k-1)$. Ordinary moments are finite linear combinations of factorial moments, so every ordinary moment converges. The Poisson distribution is moment-determinate; the method of moments gives (22). $\square$

Corollary 5.2 (The fixed-multiplicity OEIS law). *For every fixed integer $k \geq 2$,*

$$\boxed{\frac{B_{n,k}}{(kn)!} = p_{n,k} \longrightarrow e^{-(k-1)}.} \quad (23)$$

Equivalently,

$$S_{n,k} \sim \frac{(kn)!}{(k!)^n} e^{-(k-1)}. \quad (24)$$

Proof. Because the variables are integer-valued,

$$\mathbb{P}(X_{n,k} = 0) = \mathbb{P}(-\tfrac{1}{2} < X_{n,k} < \tfrac{1}{2}).$$

The endpoints $\pm \frac{1}{2}$ carry no mass under the limiting Poisson law, so weak convergence implies convergence of this probability. Hence $\mathbb{P}(X_{n,k} = 0) \rightarrow \mathbb{P}(\text{Poisson}(k-1) = 0) = e^{-(k-1)}$. Equation (24) follows from (3). $\square$

Remark 5.3. The proof is short because the correct object is not merely the zero-event count. The entire adjacency statistic has a stable Poisson limit. The constant $k-1$ is already forced by the exact mean (5).

6 A gamma expectation and an all-orders expansion engine

The exact finite sum admits a useful probabilistic transform. Let T_N have the gamma density

$$f_N(t) = \frac{t^N e^{-t}}{N!}, \quad t > 0, \quad T_N \sim \text{Gamma}(N+1, 1). \quad (25)$$

Since

$$\mathbb{E} T_N^{-M} = \frac{(N-M)!}{N!} = \frac{1}{(N)_{\underline{M}}} \quad (0 \leq M \leq N), \quad (26)$$

we obtain the following exact formula.

Proposition 6.1 (Gamma representation). *For $N = kn$,*

$$\boxed{\Phi_{n,k}(v) = \mathbb{E} \left[R_k \left(\frac{v}{T_N} \right)^n \right]}. \quad (27)$$

The gamma variable is concentrated at scale $\sqrt{N}$ around $N+1$. Since $R_k(w) = 1 + k(k-1)w + O_k(w^2)$, the exponent in (27) tends to $(k-1)v$. This already supplies another proof of the Poisson limit. More importantly, it supports a complete Laplace expansion.

For a nonnegative integer M , define polynomials $H_r(M)$ by

$$\prod_{j=0}^{M-1} \frac{1}{1-jx} = \sum_{r \geq 0} H_r(M) x^r. \quad (28)$$

Thus $H_r(M) = h_r(0, 1, \dots, M-1)$, a complete homogeneous symmetric polynomial. The first four are

$$H_0(M) = 1, \quad (29)$$

$$H_1(M) = \frac{M(M-1)}{2}, \quad (30)$$

$$H_2(M) = \frac{M(M-1)(3M^2 + M - 2)}{24}, \quad (31)$$

$$H_3(M) = \frac{M^2(M-1)^2(M+1)(M+2)}{48}. \quad (32)$$

Let $D = v d/dv$.

Theorem 6.2 (Constructive all-orders expansion). *Fix $k \geq 2$. As $n \rightarrow \infty$, uniformly for v in every compact subset of $\mathbb{C}$, $\Phi_{n,k}(v)$ has a Poincare expansion in integer powers of $1/n$. A coefficientwise construction is*

$$\boxed{\Phi_{n,k}(v) \sim \sum_{r \geq 0} \frac{1}{N^r} H_r(D) \left[R_k \left(\frac{v}{N} \right)^n \right], \quad N = kn.} \quad (33)$$

After expanding the bracket in $1/N$, collecting total powers gives any desired correction term. In particular, every coefficient is an explicit polynomial in v and a rational function of k , multiplied by $e^{(k-1)v}$.

Proof. Write

$$R_k(z)^n = \sum_{M=0}^{n(k-1)} r_{n,M} z^M.$$

Then (20) is

$$\Phi_{n,k}(v) = \sum_M r_{n,M} \frac{v^M}{N^M} \frac{N^M}{(N)_{\underline{M}}}.$$

For fixed M ,

$$\frac{N^M}{(N)_{\underline{M}}} = \prod_{j=0}^{M-1} \left(1 - \frac{j}{N}\right)^{-1} \sim \sum_{r \geq 0} \frac{H_r(M)}{N^r}.$$

Since D acts on v^M by multiplication by M , this is precisely the formal operator rule (33).

It remains to justify the asymptotic use when M is not fixed. The exact gamma representation (27) does so. Split the integral into $|T_N - N| \leq N^{2/3}$ and its complement. On the upper tail, gamma Chernoff bounds are enough. On the lower tail one combines the polynomial bound $|R_k(v/t)| \leq C_{k,V}(1 + t^{1-k})$ for $|v| \leq V$ with Stirling's formula; the exact negative moments show that terms whose total polynomial degree is proportional to N are in fact superexponentially small. Thus the weighted complement is smaller than every fixed power of N^{-1} . On the central interval, T_N^{-1} , $\log R_k(v/T_N)$, and the gamma density all possess Taylor expansions with remainders uniform for v in a prescribed compact set. After the change $T_N = N + \sqrt{N}y$, each coefficient is a polynomial in y times a Gaussian majorant, so termwise integration is valid to arbitrary fixed order. Odd half-powers cancel in the integrated expansion, leaving integer powers of $1/N$. The coefficientwise operator computation and the Laplace computation agree because both expand the same exact negative moments (26). This proves the Poincare expansion and the stated uniformity. $\square$

Remark 6.3 (Status of the analytic step). [Added 1 October 2026, batch 73O2.] The tail estimate is sketched. The coefficientwise operator rule is exact algebra, but the passage to an asymptotic statement uniform for v in compact sets is not proved in detail: the gamma Chernoff bounds, the superexponential smallness of terms of total degree proportional to N , and the cancellation of odd half-powers are asserted, not derived. A uniform factorial-moment bound of the type proved in the companion path-forest report **a189281-path-forest-expansions** of this collection (its Lemma 4.1, “Uniform factorial-moment bound”, label `spf:lem:moment-bound`, which bounds the k th factorial moment by $(\theta e^2)^k/k!$ for its own model) would make the truncation rigorous; that lemma is not proved here for the present model.

Accordingly, Theorem 6.2 and everything that uses it rest on this sketched step: the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ stated after Theorem 7.1, Theorem 8.1 (through “the fifth and higher factorial cumulants begin at $O_k(n^{-4})$ ”), Corollaries 8.2 and 8.3, the specializations (44)–(47), the count, ratio and inverse expansions (49), (52) and (54), and the correction terms in the proposed OEIS comments. They do not depend on it: Theorems 2.1, 3.1, 4.1, 5.1 and 7.1, Proposition 6.1, and Corollary 5.2, the fixed- k limit law that settles the conjecture on A330266. The intake's numerical check is consistent with the displayed coefficients: for $k = 2, \dots, 5$ the residual after the n^{-3} term of (42), multiplied by $e^{k-1}n^4$, is nearly the same at $n = 100$ and $n = 400$, as an $O(n^{-4})$ remainder requires. That is evidence, not a proof.

[2 October 2026, batch 77P5.] The gap is closed in Part II, for fixed k . Theorem II.3.1 proves the exact factorial-moment bound $\mathbb{E}(X_{n,k})_{\underline{M}} \leq (k-1)^M$ for the present model, the device asked for above, proved directly rather than imported from the path-forest report. Theorem II.4.2 turns it

into an explicit remainder, uniform for $|v| \leq V$; Corollary II.5.1 gives the integer-power expansion without any cancellation of half-powers; and Theorem II.6.1 proves the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$. Appendix II.A verifies the gamma-tail route of the proof above. Every item listed above as resting on the sketched step is therefore proved for fixed k (Remark II.6.3). Nothing uniform in k is proved; research question 1 of Section 14 is untouched.

[5 October 2026, batch 101.] Part III, an independent repair of the same step from the same pin, proves it again; Section III.2 maps it onto Part II. It adds a second proof of Theorem II.3.1 and the next coefficients (Theorem III.7.1).

7 Exact factorial cumulants

Define factorial cumulants $\kappa_r^{(F)}$ by

$$\log \Phi_{n,k}(v) = \sum_{r \geq 1} \kappa_r^{(F)} \frac{v^r}{r!}. \quad (34)$$

They are particularly natural for a Poisson limit: a Poisson variable of mean λ has $\kappa_1^{(F)} = \lambda$ and all higher factorial cumulants zero.

Theorem 7.1 (First four factorial cumulants). *Let $N = kn$. Whenever the displayed denominators are nonzero,*

$$\kappa_1^{(F)} = k - 1, \quad (35)$$

$$\kappa_2^{(F)} = -\frac{(k-1)^2}{N-1}, \quad (36)$$

$$\kappa_3^{(F)} = \frac{2(k-2)(k-1)^2}{(N-2)(N-1)}, \quad (37)$$

$$\kappa_4^{(F)} = -\frac{6(k-1)^2\{N(k^2-6k+7)+4k-6\}}{(N-3)(N-2)(N-1)^2}. \quad (38)$$

In particular,

$$\boxed{\mathbb{E}X_{n,k} = k - 1, \quad \text{Var } X_{n,k} = (k-1) - \frac{(k-1)^2}{kn-1}.} \quad (39)$$

Proof. Keep terms through v^4 in (20). Since

$$R_k(z) = 1 + a_{k,1}z + a_{k,2}z^2 + a_{k,3}z^3 + a_{k,4}z^4 + O(z^5),$$

ordinary multinomial expansion gives the first four factorial moments as rational functions of n and N . Taking the formal logarithm and using (14) simplifies to (35)–(38). This is finite polynomial algebra; Appendix A records a reproducible symbolic check. For any integer-valued random variable, $\mathbb{E}X = \kappa_1^{(F)}$ and $\text{Var } X = \kappa_2^{(F)} + \kappa_1^{(F)}$, which gives (39). $\square$

All higher factorial cumulants tend to zero. The all-orders theorem gives the sharper scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ for fixed $r \geq 2$, a mod-Poisson pattern that controls local corrections. [2 October 2026, batch 77P5: this scale does not follow from the existence of the expansion alone; it is proved as Theorem II.6.1 of Part II.]

8 Explicit expansion through third order

Put $\lambda = k - 1$. Expanding the exact cumulants and applying Theorem 6.2 gives the following compact logarithmic form.

Theorem 8.1 (Logarithmic factorial-PGF expansion). *For fixed $k \geq 2$, uniformly for v in compact subsets of $\mathbb{C}$,*

$$\begin{aligned} \log \Phi_{n,k}(v) = & \lambda v - \frac{\lambda^2 v^2}{2kn} \\ & + \frac{\lambda^2 v^2 \{2(k-2)v - 3\}}{6k^2 n^2} \\ & - \frac{\lambda^2 v^2 \{(k^2 - 6k + 7)v^2 - 4(k-2)v + 2\}}{4k^3 n^3} + O_{k,v}(n^{-4}). \end{aligned} \quad (40)$$

Proof. Insert (36)–(38) into (34) and expand their denominators in $1/n$. These terms supply every contribution through n^{-3} because the fifth and higher factorial cumulants begin at $O_k(n^{-4})$, as follows from Theorem 6.2. Collecting powers of v gives (40). [2 October 2026, batch 77P5: the step on the fifth and higher factorial cumulants is Theorem II.6.1 of Part II, and (70) there re-derives the three displayed coefficients from the full operator.] $\square$

Setting $v = -1$ produces the no-adjacency probability.

Corollary 8.2 (No-adjacency transseries). *For fixed $k \geq 2$,*

$$\log p_{n,k} = -(k-1) - \frac{(k-1)^2}{2kn} - \frac{(k-1)^2(2k-1)}{6k^2 n^2} - \frac{(k-1)^4}{4k^3 n^3} + O_k(n^{-4}), \quad (41)$$

$$\begin{aligned} p_{n,k} = e^{-(k-1)} & \left[1 - \frac{(k-1)^2}{2kn} + \frac{(k-1)^2(3k^2 - 14k + 7)}{24k^2 n^2} \right. \\ & \left. - \frac{(k-1)^4(k^2 - 10k + 17)}{48k^3 n^3} + O_k(n^{-4}) \right]. \end{aligned} \quad (42)$$

The logarithmic form is often numerically superior because exponentiating a truncated log preserves positivity and partially resums products of lower orders. [5 October 2026, batch 101: the n^{-4} terms of (41) and (42) are (81) and (80) of Part III divided by k^4 (Corollary III.7.2).]

8.1 A first local correction

The expansion describes more than the mass at zero.

Corollary 8.3 (Fixed local probabilities). *For each fixed $j \geq 0$,*

$$\mathbb{P}(X_{n,k} = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 - \frac{(\lambda - j)^2 - j}{2kn} + O_{k,j}(n^{-2}) \right]. \quad (43)$$

Proof. Exponentiating the first correction in (40) gives

$$\Phi_{n,k}(v) = e^{\lambda v} \left(1 - \frac{\lambda^2 v^2}{2kn} + O(n^{-2}) \right).$$

Write $v = u - 1$ and extract the coefficient of u^j . Dividing the correction by the Poisson mass gives

$$-\frac{\lambda^2}{2kn} \left(1 - \frac{2j}{\lambda} + \frac{j(j-1)}{\lambda^2} \right),$$

which is (43). □

9 Specializations to the OEIS rows

For $k = 2, 3, 4, 5$, (42) becomes

$$p_{n,2} = e^{-1} \left(1 - \frac{1}{4n} - \frac{3}{32n^2} - \frac{1}{384n^3} + O(n^{-4}) \right), \quad (44)$$

$$p_{n,3} = e^{-2} \left(1 - \frac{2}{3n} - \frac{4}{27n^2} + \frac{4}{81n^3} + O(n^{-4}) \right), \quad (45)$$

$$p_{n,4} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}) \right), \quad (46)$$

$$p_{n,5} = e^{-4} \left(1 - \frac{8}{5n} + \frac{8}{25n^2} + \frac{128}{375n^3} + O(n^{-4}) \right). \quad (47)$$

Thus (44) sharpens the known e^{-1} law for A007060 [2], (45) sharpens the e^{-2} law printed on A193624 [3], and (46) supplies explicit corrections for A330266 [1]. [5 October 2026, batch 101: the next term of (46) is $+7395/(32768n^4)$, (82); the n^{-4} coefficients for $k = 2, \dots, 6$ are tabulated in Corollary III.7.2.]

9.1 The standard 52-card deck

For $n = 13$ and $k = 4$, exact enumeration gives

$$p_{13,4} = 0.04547628233109430448968591097164118590821040881636 \dots \quad (48)$$

The successive approximations are

approximation	numerical value
e^{-3}	0.04978706836786394298
through n^{-1} in (46)	0.04547857206679879407
through n^{-2} in (46)	0.04547166742529067684
through n^{-3} in (46)	0.04547585004466578632
exponential of (41) through n^{-3}	0.04547622509184257975
exact	0.04547628233109430449

The third-order logarithmic approximation has relative error about 1.26×10^{-6} even at the modest value $n = 13$. [5 October 2026, batch 101: adding the n^{-4} term of (82) to the row “through n^{-3} ” gives 0.04547624344, error -3.9×10^{-8} (Remark III.7.4).]

The exact mean number of equal-rank adjacencies is 3, while

$$\text{Var } X_{13,4} = 3 - \frac{9}{51} = \frac{48}{17} = 2.8235294117 \dots$$

This quantifies the slight underdispersion relative to a Poisson(3) law.

9.2 Numerical convergence across several multiplicities

The following table lists $e^{k-1}p_{n,k}$, which tends to 1. The final column is the signed error after subtracting all three displayed correction terms in (42).

k	n	$e^{k-1}p_{n,k}$	third-order residual
2	10	0.97406694843814	7.05×10^{-6}
2	50	0.99496248890932	9.74×10^{-9}
3	10	0.93191894503056	1.77×10^{-5}
3	50	0.98660782736509	2.49×10^{-8}
4	10	0.88747575415541	2.56×10^{-5}
4	50	0.97749213853337	3.70×10^{-8}
5	10	0.84356657118181	2.52×10^{-5}
5	50	0.96813076805226	3.74×10^{-8}
6	10	0.80107140086354	1.29×10^{-5}
6	50	0.95869552482316	1.94×10^{-8}

The small residuals are not used as proof; they are independent diagnostics of the symbolic expansion. [5 October 2026, batch 101: they agree with the n^{-4} coefficients of Part III (Corollary III.7.2); at $n = 50$ each residual is within 4% of $\mathcal{P}_4(-1)/(kn)^4$, for example $7395/(32768 \cdot 50^4) \approx 3.61 \times 10^{-8}$ against 3.70×10^{-8} for $k = 4$.]

10 Asymptotics of the counts, ratios, and inverse functions

10.1 Logarithmic count transseries

Let $N = kn$ and $\lambda = k - 1$. Combining Stirling's expansion with (41) yields

$$\log B_{n,k} = N(\log N - 1) + \frac{1}{2} \log(2\pi N) - \lambda + \frac{A_1}{N} + \frac{A_2}{N^2} + \frac{A_3}{N^3} + O_k(N^{-4}), \quad (49)$$

where

$$A_1 = \frac{1}{12} - \frac{\lambda^2}{2}, \quad A_2 = -\frac{\lambda^2(2k-1)}{6}, \quad A_3 = -\frac{1}{360} - \frac{\lambda^4}{4}. \quad (50)$$

For multiset words,

$$\log S_{n,k} = \log B_{n,k} - n \log(k!). \quad (51)$$

These formulas separate the universal factorial growth from the finite adjacency penalty.

10.2 Successive-term ratios

For fixed k ,

$$\frac{B_{n+1,k}}{B_{n,k}} = \prod_{j=1}^k (kn + j) \left[1 + \frac{(k-1)^2}{2kn^2} + \frac{(k-1)^2(k-2)}{6k^2n^3} + O_k(n^{-4}) \right]. \quad (52)$$

Indeed, the factorial ratio is exact and subtraction of $\log p_{n,k}$ from $\log p_{n+1,k}$ gives the bracket. Such a ratio is useful for checking recurrences attached to individual OEIS rows.

10.3 A Lambert- W inverse transseries

Suppose a large value Y is known and one wants to estimate the index n for which $B_{n,k} \approx Y$. Set

$$L = \log Y + \lambda, \quad N_0 = \frac{L}{W_0(L/e)}. \quad (53)$$

Then $N_0(\log N_0 - 1) = L$. One Newton correction from the remaining terms in (49) gives

$$N = N_0 - \frac{\frac{1}{2} \log(2\pi N_0) + A_1/N_0 + A_2/N_0^2 + A_3/N_0^3}{\log N_0} + O_k(N_0^{-1}), \quad n = \frac{N}{k}. \quad (54)$$

This is an inverse transseries in the same spirit as the inversion examples in the ProveIt repository. Further Newton steps and further terms of Theorem 6.2 extend it mechanically. [Added 1 October 2026, batch 73O2: the inversion examples meant are those of the canonical volume *Transseries and Inversion* [15]. The forward map $\log B_{n,k}$ of (49) is the volume’s gamma carrier (theorem “All-orders expansion of Γ_+^{-1} ”, label `p6:thm:gamma`) perturbed by the power series in $1/N$ of (41), the situation of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`); the volume’s chapter “The subfactorial” (`p8:sec:top`) treats the derangement numbers the same way. Normalizations differ: here $L = \log Y + \lambda$ and $N_0 = L/W_0(L/e)$, with $\frac{1}{2} \log(2\pi N_0)$ left in the correction; the volume uses $R = \log y - \frac{1}{2} \log(2\pi)$. Formula (54) is a first-order instance of that apparatus, and no novelty is claimed for it. Deciding the integer index from an approximate inverse, as in the deck example below and in research question 11 of Section 14, is the subject of the volume’s staircase theorem (`p0:thm:staircase`).] [2 October 2026, batch 77P5: retaining A_2, A_3 in a single Newton step does not by itself give the corresponding extra accuracy, since omitted quadratic Newton terms can be larger. Part II proves the sharper error $O_k(1/(N_0 \log N_0))$ along the exact count sequence, (71), which implies the error stated in (54).] [5 October 2026, batch 101: Part III adds the same estimate for the inverse of the smooth carrier at arbitrary Y (Remark III.8.1), and the next count constant A_4 (Remark III.7.3); neither certifies an integer index.]

For the exact standard-deck count in (48), with $k = 4$, the leading Lambert estimate gives $n \approx 13.17720934$. The correction (54) gives

$$n \approx 12.99979618,$$

which correctly identifies the integer index 13.

11 Relation to rook polynomials and recent multipartite work

The coefficients $a_{k,m} = m! \binom{k}{m} \binom{k-1}{m}$ are the familiar rook numbers of a rectangular board of dimensions k by $k-1$. This explains why rook-polynomial methods naturally occur in exact deck-shuffle enumeration. Enouen’s report on the standard deck [8] develops such a direct rook-polynomial approach. The reciprocal polynomial $C_k(z) = R_k(-z)$ packages all ranks into the compact power $C_k(z)^n$.

Balanced Smirnov words have also been connected explicitly with Hamiltonian paths and cycles in complete multipartite graphs. Recent work by Mehiri [9] gives bijections and inclusion–exclusion formulas, including nonuniform part sizes. The present asymptotic regime keeps the part size k fixed and lets the number of parts n grow. In this regime, Theorem 5.1 says:

Among all ordered vertex lists of $K_{k,k,\dots,k}$, the fraction that form Hamiltonian paths tends to $e^{-(k-1)}$, and the number of forbidden within-part consecutive steps tends to Poisson($k-1$).

This probabilistic statement complements exact Hamiltonian enumerations and suggests analogous cyclic and nonuniform limit laws.

12 Reproducibility and a formalization route

The accompanying program `code/verify.py` performs the following tasks using exact Python integers for enumeration and high-precision decimal arithmetic for probabilities:

1. constructs $C_k(z) = \sum_m (-1)^m a_{k,m} z^m$;
2. raises it to the n th power by exact polynomial convolution;
3. evaluates (18);
4. checks the initial terms of A007060, A193624, and A330266;
5. verifies the exact 52-card value;
6. writes the numerical convergence table used above.

The optional SymPy script `code/derive_symbolic.py` reconstructs the first four factorial cumulants and checks the displayed n^{-3} expansion. The article does not rely on a fitted numerical constant.

A Lean formalization can be separated into finite and analytic layers. The finite core is especially approachable:

1. define rank words, runs, and the statistic X ;
2. prove the run-compression bijection and finite coefficient identity (8);
3. prove the polynomial identity (11);
4. reduce the integral notation to the factorial moment identity $\int_0^\infty t^m e^{-t} dt = m!$;
5. prove the exact finite sum (18);
6. establish the leading factorial-moment estimate using polynomial degree bounds.

The final method-of-moments step can either use an existing probability convergence library or be replaced by direct convergence of each fixed mass through the probability generating function. The exact count identities can be formalized independently of the asymptotic layer.

13 Proposed OEIS updates

Subject to independent review and attribution, the following concise comments would strengthen the relevant entries.

General fixed- k family. If $B_{n,k}$ counts permutations of kn labeled cards, k in each of n ranks, with no equal adjacent ranks, then for each fixed $k \geq 2$,

$$\frac{B_{n,k}}{(kn)!} = e^{-(k-1)} \left(1 - \frac{(k-1)^2}{2kn} + O_k(n^{-2}) \right).$$

More precisely, the number of equal-rank adjacencies converges to Poisson($k-1$), and (42) gives three correction terms.

A330266. Add (46), together with the exact variance $3 - 9/(4n-1)$ for the adjacency statistic. The integral on the entry is the $k=4$ instance of

$$B_{n,k} = \int_0^\infty \{(-1)^k k! L_k^{(-1)}(t)\}^n e^{-t} dt.$$

A007060 and A193624. Add (44) and (45), respectively. Their multiset partners A114938 and A193638 have the same normalized probabilities because of (3).

A plain-text version is included as `oeis_proposed_updates.txt`; it is explicitly marked as unsubmitted. [Added 1 October 2026, batch 73O2: shipped as `data/oeis_proposed_updates.txt`, unchanged. Nothing has been submitted to the OEIS. Its correction terms rest on the sketched step of Remark 6.3.] [2 October 2026, batch 77P5: that step is now proved for fixed k in Part II (Remark II.6.3); the comments remain unsubmitted.] [5 October 2026, batch 101: the n^{-4} terms of Part III (Corollary III.7.2) were not added to the shipped draft, which is unchanged and unsubmitted.]

14 Further research questions and topics

1. **Uniformly growing multiplicity.** Determine the largest range $k = k(n)$ in which a Poisson approximation remains valid, and locate transitions to compound-Poisson or normal behavior. The fixed- k proof is not uniform when the degree of R_k grows. [5 October 2026, batch 101: still open; Part III's question 1 records why the k -free bound of Theorem II.3.1 does not suffice.]
2. **Explicit total-variation bounds.** Use Stein–Chen coupling or the gamma representation to obtain a sharp bound of the form $d_{TV}(X_{n,k}, \text{Poisson}(k-1)) \leq C_k/n$, preferably with the optimal constant and a computable second-order signed correction.
3. **Cyclic words and Hamiltonian cycles.** Also forbid equality between the last and first ranks. Find the exact analogue of (40), taking proper account of oriented versus unoriented cycles and rotational symmetries.
4. **Unequal multiplicities.** For ranks of sizes $k_1, \dots, k_n$ and total N , identify natural sparse conditions under which the adjacency count is asymptotically Poisson with mean $\sum_i k_i(k_i-1)/N$. Determine the correction terms in symmetric sums of the k_i . [5 October 2026, batch 101: advanced. Part III proves the exact factorial-moment identity and the mean $\sum_i k_i(k_i-1)/N$ (Proposition III.3.1), the bound $\mathbb{E}(X)_{\underline{M}} \leq (k_{\max} - 1)^M$ (Proposition III.4.2), and the Poisson limit when the k_i stay bounded and the mean converges (Proposition III.4.3). Unbounded multiplicities and the correction terms remain open (question 3 of Section III.9).]

5. **Larger exclusion distance.** Forbid two equal ranks within distance d rather than only distance one. Cluster expansions should produce a Poisson or compound-Poisson law when k, d are fixed, but overlapping short patterns require new local cumulants.
6. **Moderate and large deviations.** Analyze $\mathbb{P}(X_{n,k} \geq x)$ outside the fixed- x regime. The exact gamma expectation may lead to a saddle-point approximation uniformly for x growing with n .
7. **All-order closed forms.** The operator rule (33) is constructive. Find a canonical Bell-polynomial expression for the coefficient of n^{-r} in $\log \Phi_{n,k}(v)$, and study sign or factorization patterns in k and v . [2 October 2026, batch 77P5: re-scoped. Part II gives the coefficients by the finite formula (66) and proves that the coefficient $q_s(v)$ of N^{-s} has degree at most $s + 1$ in v (Theorem II.6.1); a canonical Bell-polynomial closed form and the sign and factorization patterns remain open.] [5 October 2026, batch 101: q_4 and $\mathcal{P}_4$ are now explicit, (78)–(79); the closed form is still open.]
8. **Laguerre-zero geometry.** Clarify how the real zeros of $L_k^{(-1)}$ govern recurrences, cancellations, and possible saddle points of the exact integral. The integral involves a sign-changing polynomial, so a naive positive Laplace principle is insufficient.
9. **Differential equations and recurrences.** For each fixed k , derive minimal-order polynomial recurrences for $B_{n,k}$ and compare their asymptotic solutions with (42). Seek bounds on recurrence order as a function of k .
10. **New OEIS rows and arrays.** Create a two-dimensional array for $S_{n,k}$ or $B_{n,k}$, with rows indexed by k , including $k = 5, 6$ and exact kernels P_k . Cross-reference the labeled, multiset, rook-polynomial, and multipartite-graph interpretations.
11. **Rigorous inverse error bounds.** Turn (54) into an explicit bracketing theorem for the least n with $B_{n,k} \geq Y$, including monotonicity thresholds and certified rounding. [2 October 2026, batch 77P5: re-scoped. Part II proves the error $O_k(1/(N_0 \log N_0))$ along the exact count sequence (71), and states explicitly that this certifies no finite integer index; the bracketing theorem asked for here remains open.] [5 October 2026, batch 101: Part III adds the inverse of the smooth carrier for arbitrary Y (Remark III.8.1); the bracketing theorem remains open.]
12. **Machine-checked asymptotics.** Formalize the finite identities in Lean, then isolate a reusable theorem that turns fixed factorial-moment convergence plus tightness into Poisson convergence. This would support many related OEIS limit laws.
13. **Beyond all algebraic orders.** [Added 5 October 2026, batch 101.] Decide whether the fixed- k series in $1/n$ converges or diverges, how its coefficients grow, and whether an optimal truncation with an exponentially improved remainder exists; question 2 of Section III.9 (Part III, from bundle Report 98).

15 Conclusion

The constant $e^{-(k-1)}$ is not an isolated feature of three OEIS rows. It is the zero mass of a limiting Poisson law for the complete adjacency statistic. The run transform converts the combinatorics into a single Laguerre polynomial; reciprocal coefficients give exact inclusion–exclusion; and a gamma

negative-moment representation produces systematic correction terms. The resulting framework simultaneously proves the fixed- k conjecture recorded on A330266, sharpens the known $k = 2, 3, 4$ asymptotics, and opens a tractable family of cyclic, nonuniform, and growing-multiplicity problems.

A Cumulant algebra

For completeness, write

$$R_k(z) = 1 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 + O(z^5),$$

where

$$\begin{aligned} a_1 &= k(k-1), \\ a_2 &= \frac{k(k-2)(k-1)^2}{2}, \\ a_3 &= \frac{k(k-3)(k-2)^2(k-1)^2}{6}, \\ a_4 &= \frac{k(k-4)(k-3)^2(k-2)^2(k-1)^2}{24}. \end{aligned}$$

The coefficient of v^r in (20) is $[z^r]R_k(z)^n/(N)_{\underline{r}}$. Thus, for example,

$$\begin{aligned} \mathbb{E}(X)_{\underline{2}} &= \frac{2na_2 + n(n-1)a_1^2}{(N)_{\underline{2}}}, \\ \mathbb{E}(X)_{\underline{3}} &= \frac{6na_3 + 6n(n-1)a_2a_1 + n(n-1)(n-2)a_1^3}{(N)_{\underline{3}}}. \end{aligned}$$

The fourth factorial moment is obtained from the five partitions $4, 3+1, 2+2, 2+1+1$, and $1+1+1+1$. Applying the standard moment-to-cumulant relations and simplifying gives (35)–(38). The accompanying SymPy script performs exactly this finite calculation and verifies that the difference from (40) is $O(n^{-4})$.

B Exact initial terms for $k = 5$

The same formula supplies a useful prospective OEIS row. For $n = 0, 1, \dots, 7$,

$$\begin{aligned} B_{n,5} = & 1, \ 0, \ 28800, \ 12420864000, \ 27917203599360000, \\ & 1977269653324800000000000, \\ & 3614614673967242167910400000000, \\ & 147444323126376251733903684403200000000. \end{aligned}$$

The kernel is $P_5(t) = t^5 - 20t^4 + 120t^3 - 240t^2 + 120t$, and $B_{n,5} = \int_0^\infty P_5(t)^n e^{-t} dt$. These terms should be independently cross-checked before creating or editing an OEIS entry.

C Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

Source. The report was built from one manuscript, manuscript 57 of batch 73 of ProveIt’s incoming-reports intake: the archive `ProveIt_Balanced_Smirnov_OEIS.zip` (main file `article.tex`, a 17-page PDF). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision and cites no repository file. It has no author or preparer line: its author field is the subtitle “A research note on OEIS A007060, A193624, A330266, and related sequences”, and the delivered PDF’s author metadata is empty.

What the write changed.

- Every label gained the prefix `bsw::`; none was otherwise renamed or removed. The manuscript had 66 labels.
- Citation commands were added: the manuscript listed twelve bibliography entries but cited none of them. They now stand at the first mentions of the OEIS entries, the Smirnov-word generating function, the Laguerre expansion, and the works of Eriksson–Martin, Enouen and Mehiri.
- The authors and dates of the two OEIS contributions are named: S. Kirgizov’s fixed- k conjecture (29 September 2023) and D. Radcliffe’s deduction of the case $k = 4$ (9 September 2025).
- Added: the editorial note after the scope paragraph, Remark 6.3 on the sketched analytic step, a note in the inverse section citing the transseries volume [15], a note on the shipped file of proposed OEIS comments, one bibliography entry, the label `bsw:sec:questions`, and this section.
- Nothing was removed. Every non-claim of the manuscript is kept: no complete literature search or priority claim, the numerical residuals “not used as proof”, the cross-check required for the $k = 5$ terms, and the unsubmitted status of the proposed OEIS comments.

Relation to the companion report. The path-forest report `a189281-path-forest-expansions` of this collection (batch 73, manuscript 60) follows the same chain of method (exact marked-subset generating function, Poisson limit, all-orders $1/n$ expansion of a factorial probability generating function, Lambert- W inversion) for permutations avoiding a fixed positional and value offset, with limit $\text{Poisson}(\theta)$, $\theta \in \{1, 2\}$. Neither theorem specializes to the other. In the present model the n ranks give n cliques K_k as the “value” relation, whose number grows with n , so the hypotheses of the path-forest report (a fixed number of long paths) do not hold. Its uniform factorial-moment bound is the device named in Remark 6.3.

Checks by the intake. The intake reran both shipped programs on a copy (the CSV is byte-identical; the deck output equal apart from line endings; the symbolic script confirms $\kappa_1^{(F)}, \dots, \kappa_4^{(F)}$ and the n^{-3} expansion), compared the terms of A007060, A193624 and A330266 with an independent dynamic program over rank words, confirmed the $k = 5$ terms (for $n \leq 4$ by that program, for $n \leq 7$ by the formula) and the 52-card value, checked the variances $4/5, 6/5, 6/7$ at $(n, k) = (3, 2), (2, 3), (4, 2)$ by full enumeration, and rederived by hand the local correction, the ratio bracket and A_1, A_2, A_3 . These checks do not replace an independent review of the proofs.

Part II. [Added 2 October 2026, batch 77P5.] Part II was added from a second manuscript, manuscript 11 of batch 77, which pins this report at commit `4b874cea0`; its provenance, the choices of the merge and the intake’s checks are in Section II.9. Besides Part II, the batch-77 write added the dated notes of 2 October 2026 in this Part and two bibliography entries (Gessel, Taylor), and changed no earlier text.

Part III. [Added 5 October 2026, batch 101.] Part III was added from bundle Report 98 of batch 101, which pins this report at `4b874cea0`, as Part II does; its provenance, the choices of the merge and the intake’s checks are in Section III.11. Besides Part III, the batch-101 write added the notes dated 5 October 2026 in Parts I and II and research question 13, and changed no earlier text.

Part II

A rigorous uniform remainder for balanced Smirnov asymptotics

Editorial note (2 October 2026, batch 77P5). Part II is manuscript 11 of batch 77 of ProveIt’s incoming-reports intake, *A rigorous uniform remainder for balanced Smirnov asymptotics*, dated 2 October 2026, whose author line reads “Mathematical repair and reproducibility report”. It arrived in the archive `balanced-smirnov-repair-reproducibility.zip`; AI-assisted, unrefereed, not formalized. It audits this report as it stood at commit `4b874cea0` and proves the analytic step that Remark 6.3 records as a sketch. Sections 1–15 and Appendices A–C above form Part I, unchanged except for notes dated 2 October 2026. The write prefixed every label of the manuscript with `bsw:rp:`, replaced its citations of “Theorem 6.2”, “Remark 6.3” and “Theorem 4.1” of the pinned source by cross-references, renamed the symbols listed in Section II.1, printed once the formulas that Part I already displays (Section II.7), and collected the source pin and the provenance in Section II.9. [5 October 2026, batch 101.] Part III is a second, independent answer to the same assignment; Section III.2 maps it onto this Part.

Summary (the manuscript’s abstract). For a uniform shuffle of n ranks of k cards, with fixed $k \geq 2$, let X count equal-rank adjacencies and put $N = kn$. We prove the exact bound $\mathbb{E}(X)_{\underline{M}} \leq (k-1)^M$ and use it to justify the full factorial-probability-generating-function expansion uniformly on compact subsets of the complex plane. The remainder has an explicit Poisson-weighted bound. A separate formal-operator argument proves the degree cancellations needed for all-orders factorial-cumulant scaling. This closes the analytic gap in Theorem 6.2 of Part I. We verify its displayed third-order formulas, give the local, count and ratio consequences, and state the inverse approximation with its necessary scope. The existing combinatorial formulas and displayed coefficients are credited to their sources; no claim of historical novelty is made.

II.1 Notation of Part II

[Added at the write, 2 October 2026.] Part II keeps Part I’s notation wherever the two agree: $N = kn$, $\lambda = k-1$, $a_{k,m}$ and $R_k(z)$ of (14)–(15), $(x)_{\underline{M}}$, $p_{n,k}$, $B_{n,k}$, $S_{n,k}$, $H_r(M)$ of (28), $D = v d/dv$, the factorial cumulants $\kappa_r^{(F)}$ of (34), A_1, A_2, A_3 of (50), and $L = \log Y + \lambda$, N_0 of (53). The manuscript wrote Φ_n for Part I’s $\Phi_{n,k}$ of (19) and e for $\mathbf{e}$; Part II uses Part I’s forms. The other symbols of the manuscript that collide with Part I, or with each other, were renamed as follows.

Part II	manuscript	meaning, and the reading to avoid
$c_{n,M}$	$c_{n,M}$	$\mathbb{E} \binom{X}{M} = r_{n,M}/(N)_{\underline{M}}$, with $r_{n,M} = [z^M]R_k(z)^n$ as in the proof of Theorem 6.2; not the carrier constant c_k of Section II.7.3
d	d	$n(k-1)$, the largest possible value of X ; not the d of d/dv
$\sigma_j(M)$	$p_j(M)$	the power sum $\sum_{i \leq M} i^j$; not the probability $p_{n,k}$
$F_N(v)$	$F_N(v)$	$R_k(v/N)^n$, the bracket in (33); not the gamma density $f_N(t)$ of (25)
$\mathcal{P}_s(v)$	$P_s(v)$	the coefficient polynomials of $e^{-\lambda v}\Phi_{n,k}(v)$ in powers of N^{-1} ; <i>not</i> the Laguerre kernels $P_k(t)$ and $P_{k,u}(t)$ of (9)–(13): $\mathcal{P}_2(v) \neq P_2(t) = t^2 - 2t$
J	L	the truncation order; L stays $\log Y + \lambda$ as in (53)
$\mathcal{C}_{k,V,J}$	$C_{k,V,L}$	the remainder constant of Theorem II.4.2; not the reciprocal polynomial $C_k(z)$ of (16)
$\mathcal{T}_j(a)$	$\mathcal{T}_j(a)$	the j th Touchard polynomial; not the gamma variable T_N
$\mathcal{H}_w$	T_w	the formal operator $\sum_r H_r(D)w^r$; at $w = 1/N$ it is an operator, <i>not</i> the gamma variable T_N of (25)
$\mathcal{A}_w(D)$	$A_w(D)$	the generator $\sum_{j \geq 1} w^j \sigma_j(D)/j$; not the Stirling constants A_1, A_2, A_3
$\Sigma(t, w, v), \Sigma_s(t, v)$	S, S_s	the formal logarithm along the operator flow; not the multiset count $S_{n,k}$
$\mathcal{B}_a$	B_a	$e^{-\Sigma} D^a e^{\Sigma}$; not the count $B_{n,k}$
γ_j	c_j	$(b_{j+1}/k)v^{j+1}$, an abbreviation in Section II.7
$\psi(x), g(x)$	$f(x), h(x)$	the inverse carrier $x(\log x - 1)$ and its correction; renamed because f_s and h_s denote other objects
M_N	J	the cut $\lfloor N^{1/3} \rfloor$ of Appendix II.A

Normalization: the manuscript expands in powers of $N^{-1} = (kn)^{-1}$, Part I in powers of n^{-1} ; a coefficient of N^{-s} is k^{-s} times the corresponding coefficient of n^{-s} . Two readings are excluded explicitly below: Theorem II.3.1 is factorial-moment domination, not stochastic domination by a Poisson variable; and $\kappa_r^{(F)}$ are *factorial* cumulants, not ordinary cumulants (Remark II.6.2).

II.2 Scope and the gap being repaired

Part I considers *fixed multiplicity*: k is fixed while the number n of ranks tends to infinity. Its Theorem 6.2 gives a formal coefficientwise operator expansion, but the transition to compact-complex-uniform asymptotics sketches three steps: weighted gamma-tail control, suppression of large coefficient degrees, and cancellation of odd half-powers. Its editorial Remark 6.3 explicitly identifies this gap. The exact generating functions, exact first four factorial cumulants, Poisson limit and leading no-adjacency limit do not require that step.

The proof below supplies a direct quantitative remainder, avoiding a saddle expansion. It also supplies a separate all-orders logarithmic degree argument: existence of a uniform expansion alone would not prove the asserted factorial-cumulant scaling. Appendix II.A additionally verifies the weighted gamma-tail assertion. No long-path hypothesis from the neighboring path-forest report [a189281-path-forest-expansions](#) is used; its theorem cannot simply be imported into this growing collection of rank cliques.

All asymptotic statements below fix k . Constants may depend on k , the compact set and the truncation order. Nothing here asserts uniformity for growing k , or a uniform relative local

approximation for growing adjacency count.

II.3 Exact model and factorial-moment domination

There are $n \geq 1$ ranks, each with $k \geq 2$ distinct cards. Choose a permutation of the $N = kn$ cards uniformly. Let $X = X_{n,k}$ be the number of positions $j < N$ at which cards $j, j+1$ have the same rank. Equivalently choose a multiset word uniformly with k copies of each of n letters. Write

$$\lambda = k - 1, \quad d = n(k - 1), \quad \Phi_{n,k}(v) = \mathbb{E}(1 + v)^X, \quad p_{n,k} = \mathbb{P}(X = 0).$$

The labeled no-adjacency count is $B_{n,k} = N!p_{n,k}$; the multiset count is $S_{n,k} = B_{n,k}/(k!)^n$. We have $0 \leq X \leq d < N$. For $k = 2, 3, 4$, respectively, the labeled counts are OEIS A007060, A193624 and A330266 [2, 3, 1]; the corresponding multiset-word counts are A114938, A193638 and A321633 [4, 5, 6]. The probability is common to both models, but the counts are different.

With $(a)_0 = 1$, $a_{k,m}$ and R_k as in (14)–(15) (the manuscript writes $a_{k,m} = \binom{k-1}{m}(k)_{\underline{m}}$, the same number), put

$$r_{n,M} = [z^M]R_k(z)^n, \quad c_{n,M} = \mathbb{E}\binom{X}{M} = \frac{r_{n,M}}{(N)_{\underline{M}}} \quad (0 \leq M \leq d). \quad (55)$$

Set $r_{n,M} = c_{n,M} = 0$ for $M > d$; do not evaluate the quotient outside its stated support. The exact identity

$$\Phi_{n,k}(v) = \sum_{M=0}^d \frac{[z^M]R_k(z)^n}{(N)_{\underline{M}}} v^M$$

is (20) of Theorem 4.1, which Part I obtains by weighted run transforms.

Second proof of (20) (manuscript 11). For completeness, there is a direct contraction proof. Canonically number the occurrences of each letter from left to right. The $k - 1$ possible bonds for a letter join consecutive occurrences; a bond is present when those occurrences are adjacent. Select m_i bonds for letter i and contract them. The contracted words have $k - m_i$ copies of letter i , and expansion according to the selected components is inverse to contraction. If $M = \sum_i m_i$, the number of such words is $(N - M)! / \prod_i (k - m_i)!$. Double counting words with M selected present bonds and dividing by $N!/(k!)^n$ gives (55). $\square$

The Laguerre and rook-polynomial machinery is older: Gessel's linear-permutation construction [10] (its Section 6 treats linear permutations), Taylor's Theorem 2.1 and run generating function [11] (his Theorem 2.1 credits Gessel's exact Carlitz-word formula; his Lemma 2.2 gives the classical generating function), and Enouen's standard-deck application [8] (its Sections 3–4 give the rook-polynomial solution and the exact standard-deck count) give relevant precedents. Eriksson and Martin [7] enumerate the same Carlitz multipermutation family. This Part repairs a specified proof, rather than claiming priority for those formulas.

Theorem II.3.1 (Exact Poisson factorial domination). *For all $n \geq 1$ and integers $M \geq 0$,*

$$0 \leq c_{n,M} \leq \frac{\lambda^M}{M!}, \quad \text{equivalently} \quad \mathbb{E}(X)_{\underline{M}} \leq \lambda^M. \quad (56)$$

Consequently $|\Phi_{n,k}(v)| \leq e^{\lambda|v|}$ for every complex v .

Proof. Since $(k)_{\underline{m}} \leq k^m$, nonnegative coefficients give

$$R_k(z) \preceq (1 + kz)^{k-1}, \quad r_{n,M} \leq k^M \binom{d}{M}.$$

Here $\preceq$ means coefficientwise inequality. For $0 \leq M \leq d$,

$$c_{n,M} \leq \frac{k^M}{M!} \prod_{j=0}^{M-1} \frac{d-j}{N-j} \leq \frac{k^M}{M!} \left(\frac{d}{N}\right)^M = \frac{\lambda^M}{M!},$$

since $(d-j)/(N-j) \leq d/N$. Empty products include $M = 0$, and the zero convention includes $M > d$. Sum the absolute coefficients to obtain the last assertion. $\square$

Remark II.3.2. This is factorial-moment domination; it is not a claim of stochastic domination by a Poisson random variable. It includes $n = 1$, when $X = k - 1$ deterministically. [Added at the write: it is the analogue, for the present model, of the uniform factorial-moment bound of the path-forest report that Remark 6.3 names, proved here directly.] [5 October 2026, batch 101: Part III gives a second proof through the majorant $(1 + \lambda z)^k$, and the analogous bound for unequal rank sizes, Proposition III.4.2.]

II.4 The explicit uniform operator remainder

For a finite list of nonnegative variables let h_s denote the complete homogeneous symmetric polynomial. Set $h_0 = 1$; for the empty list set $h_s = 0$ when $s > 0$. For integers $M, r \geq 0$ define

$$H_r(M) = h_r(0, 1, \dots, M-1), \quad Q_M(x) = \prod_{j=0}^{M-1} (1 - jx)^{-1},$$

in agreement with (28). The list for $M = 0$ is empty. Both $M = 0$ and $M = 1$ give $Q_M = 1$ and $H_r(M) = 0$ for $r \geq 1$.

Lemma II.4.1 (Complete-homogeneous remainder). *For nonnegative integers a, b , $h_{a+b} \leq h_a h_b$ on nonnegative variables. Consequently, for an integer $J \geq 0$ and $0 \leq x < 1/(M-1)$ when $M \geq 2$,*

$$0 \leq Q_M(x) - \sum_{r=0}^J H_r(M) x^r \leq x^{J+1} H_{J+1}(M) Q_M(x). \quad (57)$$

The inequality holds without restriction on $x \geq 0$ for $M = 0, 1$. For every integer $s \geq 1$,

$$H_s(M) \leq \left(\frac{M(M-1)}{2}\right)^s \leq M^{2s}. \quad (58)$$

Proof. A term of h_{a+b} is indexed by a nondecreasing string of $a + b$ indices. Splitting after a entries injects its terms, with unchanged weights, into the pairs of strings counted by $h_a h_b$. Thus $H_{J+1+t}(M) \leq H_{J+1}(M) H_t(M)$. The convergent expansion $Q_M(x) = \sum_{r \geq 0} H_r(M) x^r$ now gives (57) by summing over $t \geq 0$. Finally, the multinomial expansion of the sum of the variables raised to s counts each weak monomial with coefficient at least one; this proves (58). $\square$

Put $F_N(v) = R_k(v/N)^n$. The symbol $H_r(D)$ denotes a polynomial in the Euler differential operator $D = v d/dv$. Indeed, if

$$\sigma_j(M) = \sum_{i=0}^{M-1} i^j,$$

then

$$\sum_{r \geq 0} H_r(M) x^r = \exp \left(\sum_{j \geq 1} \frac{x^j}{j} \sigma_j(M) \right). \quad (59)$$

Faulhaber's formula gives $\deg \sigma_j = j + 1$; coefficient extraction shows that $H_r(M)$ is a polynomial in M of degree at most $2r$, with the stated values also at $M = 0$.

Theorem II.4.2 (Quantitative compact-complex remainder). *For every integer $J \geq 0$ and real $V \geq 0$, for all $n \geq 1$,*

$$\sup_{|v| \leq V} \left| \Phi_{n,k}(v) - \sum_{r=0}^J N^{-r} H_r(D) F_N(v) \right| \leq \frac{\mathcal{C}_{k,V,J}}{N^{J+1}}, \quad (60)$$

where, with $a = \lambda V$,

$$\mathcal{C}_{k,V,J} = \sum_{M=0}^{\infty} H_{J+1}(M) \frac{a^M}{M!} \leq e^a \mathcal{T}_{2J+2}(a) < \infty. \quad (61)$$

Here $\mathcal{T}_j(a) = e^{-a} \sum_{M \geq 0} M^j a^M / M!$ is the j th Touchard polynomial. For $V = 0$ the constant is zero.

Proof. The exact finite identity (20) reads

$$\Phi_{n,k}(v) = \sum_{M=0}^d r_{n,M}(v/N)^M Q_M(1/N).$$

Every M in this sum satisfies $M \leq d < N$, so Lemma II.4.1 applies. Since $D(v^M) = Mv^M$, subtract the finite operator sum, take absolute values, and use nonnegativity to obtain

$$\begin{aligned} \text{error} &\leq N^{-J-1} \sum_{M=0}^d H_{J+1}(M) r_{n,M}(V/N)^M Q_M(1/N) \\ &= N^{-J-1} \sum_{M=0}^d H_{J+1}(M) c_{n,M} V^M \\ &\leq N^{-J-1} \mathcal{C}_{k,V,J}. \end{aligned}$$

The last step uses Theorem II.3.1. The bound (58) proves finiteness and the Touchard upper bound. In particular there is no need to control $Q_M(1/N)$ uniformly in M : its entire value converts the majorant back into an exact factorial-moment coefficient. $\square$

[5 October 2026, batch 101: keeping the factor 2^{-s} of the first inequality in (58) sharpens (61) to $\mathcal{C}_{k,V,J} \leq 2^{-(J+1)} e^a \mathcal{T}_{2J+2}(a)$; Remark III.5.1 of Part III.]

II.5 Integer powers and the normalized logarithm

Choose the analytic germ

$$\log R_k(z) = \sum_{j \geq 1} b_j z^j, \quad \log R_k(0) = 0, \quad b_1 = k\lambda.$$

For w near zero set

$$F(w, v) = \exp\left(\frac{\log R_k(wv)}{kw}\right), \quad F(0, v) = e^{\lambda v}. \quad (62)$$

On each fixed v -disk the singularity at $w = 0$ is removable and the function is jointly analytic for sufficiently small $|w|$. At $w = 1/N$ it equals $F_N(v)$ exactly, since $N/k = n$ is integral. Its expansion is

$$F(w, v) = e^{\lambda v} \sum_{s \geq 0} f_s(v) w^s, \quad f_0 = 1, \quad (63)$$

$$f_s(v) = \sum_{\substack{m_1, \dots, m_s \geq 0 \\ \sum j m_j = s}} \prod_{j=1}^s \frac{((b_{j+1}/k)v^{j+1})^{m_j}}{m_j!}. \quad (64)$$

Thus f_s is a polynomial of degree at most $2s$. Analyticity on a slightly larger v -disk and Cauchy's estimates justify differentiating each fixed-order Taylor remainder finitely many times, uniformly on $|v| \leq V$.

Corollary II.5.1 (All-orders expansion). *For each fixed integer $J \geq 0$,*

$$\Phi_{n,k}(v) = e^{\lambda v} \sum_{s=0}^J \mathcal{P}_s(v) N^{-s} + O_{k,V,J}(N^{-J-1}), \quad |v| \leq V, \quad (65)$$

where $\mathcal{P}_0 = 1$ and

$$\mathcal{P}_s(v) = e^{-\lambda v} \sum_{r=0}^s H_r(D)(e^{\lambda v} f_{s-r}(v)). \quad (66)$$

Each $\mathcal{P}_s$ is a polynomial in v , with rational-in- k coefficients.

Proof. In Theorem II.4.2, expand the r th differentiated bracket through w^{J-r} . Its multiplied remainder is $O(w^{J+1})$, uniformly on the disk. Summing finitely many such remainders gives (65). All powers are integer powers from analytic Taylor series; no half-power cancellation is assumed. $\square$

On a fixed disk, $e^{-\lambda v} \Phi_{n,k}(v) = 1 + O(N^{-1})$. Hence for sufficiently large n it lies within distance $1/2$ of 1, on a neighborhood of that disk. Define

$$\log \Phi_{n,k}(v) := \lambda v + \log(e^{-\lambda v} \Phi_{n,k}(v)), \quad (67)$$

where the second logarithm is the analytic branch near 1. This branch has value zero at $v = 0$. It has a uniform integer-power expansion. An unspecified principal logarithm on a large complex disk would not give the same statement.

II.6 The all-orders factorial-cumulant degree argument

The next argument is separate from existence of the uniform expansion. It proves the degree cancellations needed to control all factorial cumulants.

Work formally in w ; all operator expressions have finitely many contributions to any fixed coefficient. By (59),

$$\mathcal{H}_w := \sum_{r \geq 0} H_r(D) w^r = \exp \mathcal{A}_w(D), \quad \mathcal{A}_w(D) = \sum_{j \geq 1} \frac{w^j}{j} \sigma_j(D). \quad (68)$$

This may be checked on each monomial v^M using the scalar identity for $Q_M(w)$. No convergence of the infinite operator series at nonzero w is asserted.

Introduce a formal flow parameter t and write

$$e^{t\mathcal{A}_w(D)} F(w, v) = e^{\Sigma(t, w, v)}, \quad \Sigma(t, w, v) = \lambda v + \sum_{s \geq 1} \Sigma_s(t, v) w^s.$$

After removal of $e^{\lambda v}$, this is a formal w -series with polynomial coefficients in v, t and constant term 1; its formal logarithm is uniquely defined. Initially,

$$\Sigma_s(0, v) = \frac{b_{s+1}}{k} v^{s+1}.$$

Theorem II.6.1 (Logarithmic degree and factorial cumulants). *The expansion of the normalized logarithm (67) has the form*

$$\log \Phi_{n,k}(v) = \lambda v + \sum_{s=1}^J q_s(v) N^{-s} + O_{k,V,J}(N^{-J-1}), \quad \deg_v q_s \leq s+1. \quad (69)$$

Consequently, for each fixed $r \geq 2$,

$$\kappa_r^{(F)} := \frac{d^r}{dv^r} \log \Phi_{n,k}(v) \Big|_{v=0} = O_{k,r}(N^{1-r}) = O_{k,r}(n^{1-r}).$$

Also $\kappa_1^{(F)} = \mathbb{E}X = \lambda$ exactly.

Proof. The formal flow satisfies

$$\partial_t \Sigma = e^{-\Sigma} \mathcal{A}_w(D) e^{\Sigma},$$

where the conjugated operator on the right acts on 1. Put $\mathcal{B}_a = e^{-\Sigma} D^a e^{\Sigma}$. Then $\mathcal{B}_0 = 1$ and

$$\mathcal{B}_{a+1} = D\mathcal{B}_a + (D\Sigma)\mathcal{B}_a.$$

Thus each monomial of $\mathcal{B}_a$ is a product of at most a factors $D^\ell \Sigma$, $\ell \geq 1$, apart from the constant for $a = 0$.

We induct on s to prove $\deg_v \Sigma_s(t, v) \leq s+1$. To extract w^s from the j th summand of the generator, $1 \leq j \leq s$, the orders s_i of these factors sum to $s-j$. Each is strictly less than s . Because σ_j has degree at most $j+1$, there are at most $b \leq j+1$ factors. The Euler derivative does not increase polynomial degree. The induction hypotheses, including $\Sigma_0 = \lambda v$, give total degree at most

$$\sum_{i=1}^b (s_i + 1) = s - j + b \leq s + 1.$$

The zero-factor case causes no problem. Therefore $\partial_t \Sigma_s$ has this degree bound, and the initial value does too. Integration in t proves the induction. At $t = 1$, the coefficients are those of $\mathcal{H}_w F$, which Corollary II.5.1 has already identified as the analytic asymptotic expansion of $\Phi_{n,k}$. Finite logarithmic-series algebra identifies $q_s = \Sigma_s(1, v)$.

For a fixed $r \geq 2$, take $J = r - 2$ in (69) on a disk about zero. All retained polynomials have degree less than r . Cauchy's coefficient formula applied to the uniform remainder gives the claimed bound. The exact mean follows from (55) with $M = 1$. $\square$

Remark II.6.2. These are *factorial* cumulants, using derivatives in v of $\log \mathbb{E}(1 + v)^X$. Ordinary cumulants use $\log \mathbb{E} e^{tX}$ and converge to the Poisson value λ ; they do not have the displayed decay.

Remark II.6.3 (What this settles in Part I). [Added at the write.] Theorem II.4.2 and Corollary II.5.1 prove Theorem 6.2: (60) is the operator rule (33) with an explicit remainder, uniform for $|v| \leq V$. Theorem II.6.1 proves the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ stated after Theorem 7.1, and with it the step “the fifth and higher factorial cumulants begin at $O_k(n^{-4})$ ” in the proof of Theorem 8.1 ($r = 5$ gives $O_k(n^{-4})$). Hence every item that Remark 6.3 lists as resting on the sketched step is now proved, for fixed k ; nothing uniform in k is claimed.

II.7 Verified formulas and consequences

The coefficients in this section are those displayed in Part I; the contribution here is the proved remainder and independent algebraic verification. The reproducibility scripts compute them from the full operator, rather than assuming that four cumulants exhaust third order. [Write: formulas that Part I already displays are cited, not printed a second time; the manuscript reprinted them in the variable N , and they agree term by term.]

The first kernel-log coefficients are

$$\begin{aligned} b_1 &= k\lambda, & b_2 &= -k\lambda^2, \\ b_3 &= \frac{k\lambda^2(5k-6)}{3}, & b_4 &= -\frac{k\lambda^2(7k^2-18k+12)}{2}. \end{aligned}$$

If $\gamma_j = (b_{j+1}/k)v^{j+1}$, then

$$f_1 = \gamma_1, \quad f_2 = \gamma_2 + \frac{1}{2}\gamma_1^2, \quad f_3 = \gamma_3 + \gamma_1\gamma_2 + \frac{1}{6}\gamma_1^3.$$

Use $e^{-\lambda v} D(e^{\lambda v} p) = vp' + \lambda vp$ in (66); logarithmic-series algebra gives

$$q_1 = -\frac{\lambda^2 v^2}{2}, \quad q_2 = \frac{\lambda^2 v^2 \{2(k-2)v-3\}}{6}, \quad q_3 = -\frac{\lambda^2 v^2 \{(k^2-6k+7)v^2-4(k-2)v+2\}}{4}, \quad (70)$$

so that (69) with $J = 3$ and $N = kn$ is exactly (40) of Theorem 8.1, with the branch (67) and uniformity on $|v| \leq V$. This verifies the cancellations of possible v^5, v^6 terms directly. [5 October 2026, batch 101: the next coefficient q_4 is (78) of Part III.]

II.7.1 No-adjacency probability and local masses

Setting $v = -1$ gives (41) and (42) of Corollary 8.2; the verified coefficients of n^{-1}, n^{-2}, n^{-3} in $e^{k-1} p_{n,k}$ for $k = 2, 3, 4, 5$ are those of (44)–(47), which the manuscript tabulates again. For each fixed integer $j \geq 0$, Cauchy's formula on a fixed circle in u , with $v = u - 1$, gives (43) of Corollary 8.3.

Second derivation of the local correction (manuscript 11). The first correction is $-\lambda^2(u-1)^2/(2N)$ times the Poisson PGF. Dividing its u^j coefficient by the Poisson mass gives $-\lambda^2 - 2\lambda j + j(j-1)]/(2N)$, including $j = 0, 1$. $\square$

II.7.2 Counts and successive ratios

Stirling's formula and (41) yield (49), with the constants A_1, A_2, A_3 of (50). The multiset count satisfies $\log S_{n,k} = \log B_{n,k} - n \log(k!)$ exactly (51). In particular its leading growth is $N(\log N - 1 - \log(k!)/k)$, rather than the labeled carrier in (49). For fixed k the ratio expansion (52) holds.

Second proof of (52) (manuscript 11). To see this without differentiating a discrete remainder, if $\log p_{n,k} = -\lambda + a/n + b/n^2 + c/n^3 + O(n^{-4})$, subtraction at consecutive integers gives $-a/n^2 + (a-2b)/n^3 + O(n^{-4})$. Exponentiation does not change those two terms, and the factorial ratio is exact. $\square$

II.7.3 Carefully scoped Lambert inverse

This subsection inverts the labeled count $B_{n,k}$, not the multiset count $S_{n,k}$. For $Y = B_{n,k}$ along the exact count sequence, define L and N_0 as in (53):

$$L = \log Y + \lambda, \quad N_0 = \frac{L}{W_0(L/e)}.$$

Thus $N_0(\log N_0 - 1) = L$. With

$$g(x) = \frac{1}{2} \log(2\pi x) + A_1/x + A_2/x^2 + A_3/x^3,$$

the following holds as $n \rightarrow \infty$:

$$N = N_0 - \frac{g(N_0)}{\log N_0} + O_k\left(\frac{1}{N_0 \log N_0}\right), \quad n = N/k. \quad (71)$$

In particular it implies the coarser $O_k(N_0^{-1})$ error of (54). Put $\psi(x) = x(\log x - 1)$. Equation (49) says $\psi(N) + g(N) = \psi(N_0) + O(N^{-4})$. First $N \sim N_0$; then $\psi' = \log x$ and $g(N) = O(\log N)$ give $N - N_0 = O(1)$. Taylor expansion of these explicit smooth functions, using $\psi''(x) = 1/x$ and $g'(x) = O(1/x)$, gives

$$(N - N_0) \log N_0 + g(N_0) = O(1/N_0),$$

which proves (71) without differentiating the discrete error.

For arbitrary Y off the sequence, a continuous interpolation or an integer-threshold convention must be specified. Asymptotic notation alone does not certify recovery of a particular finite integer index. Retaining A_2, A_3 in a single Newton correction does not by itself yield the corresponding extra inversion accuracy: omitted quadratic Newton terms can be larger. For a multiset-count inverse, the leading carrier instead uses $c_k = 1 + \log(k!)/k$ and $N_0 = L/W_0(Le^{-c_k})$, with Newton denominator $\log N_0 - \log(k!)/k$. The same proof applies after making those replacements. The inversion apparatus itself is not claimed to be new. [Added at the write: as the note in Part I's inverse section says, the forward map is the gamma carrier of the transseries volume [15] (p6:thm:gamma) perturbed by a power series in $1/N$, the setting of its theorem "Perturbed inversion" (p0:thm:perturbed-inversion); the carrier equation $\psi(N_0) = L$ is solved by W_0 directly rather

than through the dominant-block form $aX + b \log X = L$ of `p0:thm:lambert-core`. The interpolation and integer-threshold conventions asked for above are the subject of its staircase theorem (`p0:thm:staircase`). No novelty is claimed for (71).]

II.8 Reproducibility and review

[Rewritten at the write to name the shipped files.] The delivered archive contained the editable $\text{\LaTeX}$ source and a 10-page PDF of this Part, two independent Python checks, their exact outputs, a mathematical audit, provenance, replay instructions and a clean replay script. It does not redistribute the source manuscript (Part I) or third-party papers; pinned links and hashes are provided instead. This report ships the following delivered files.

shipped file	content
<code>code/02-repair-check_operator.py</code>	the operator check
<code>code/02-repair-check_independent.py</code>	the independent check
<code>code/02-repair-replay.sh</code>	the replay script
<code>data/02-repair-operator.txt</code>	recorded output of the operator check
<code>data/02-repair-independent.txt</code>	recorded output of the independent check
<code>data/02-repair-clean-replay.txt</code>	record of the delivery's clean replay
<code>data/02-repair-environment.txt</code>	the delivery's software environment
<code>data/02-repair-requirements.txt</code>	the SymPy pin
<code>data/02-repair-sources.json</code>	source pin, checksums, literature links
<code>02-repair-mathematical-audit.md</code>	the mathematical audit
<code>02-repair-integrated-review.md</code>	the integrated review
<code>02-repair-visual-qa.md</code>	visual check of the delivered PDF

The manuscript source and PDF (merged here as Part II), the PDF build script, its build log and the checksum ledger are not shipped; the README says how to rerun the checks.

The operator check constructs b_j , f_s , the complete operators $H_r(D)$ and the logarithm through order N^{-3} , and asserts the displayed identities. The independent check enumerates adjacency histograms by exact multiset-word dynamic programming for $k = 2, \dots, 5$, $n = 1, \dots, 5$; verifies every factorial moment through two indices past support; checks the remainder inequalities with rational arithmetic; and independently derives the symbolic coefficients. All assertions passed with SymPy 1.14.0. Finite checks support reproducibility but are not substitutes for the proofs above.

The accompanying mathematical audit found no substantive defect in the uniform remainder, formal degree argument, dependent formulas or optional gamma-tail proof. This is an AI-assisted, independently cross-checked mathematical report, not a peer-reviewed publication or a formal proof-assistant development. The targeted literature check is not exhaustive and establishes no historical novelty.

II.9 Provenance of Part II

[Added 2 October 2026, batch 77P5; the first paragraph is the manuscript's "Source pin" appendix.]

Source pin. The audited ProveIt commit is

4b874cea0012c51a6841ad9c58f6e20fa57c71da.

The report directory is that of this report. The provenance file `data/02-repair-sources.json` gives its complete repository path, full pinned URLs and SHA-256 checksums for the inspected `article.tex` and editorial `README.md`. The full third-party source files are not bundled. [Write: those checksums identify the two files as they stood at `4b874cea0`, which is also their state after the batch-73O2 write `728773045`; this write changed both files, so the checksums no longer match the current ones.]

Intake record. Manuscript 11 of batch 77 (cluster P5), from the archive named in the editorial note at the start of this Part; arrival commit `096ee7b87`; placement commit `4f11bc9c0`, which staged twelve delivered files with the prefix `02-repair-` and removed the archive from `docs/incoming` (it survives in `096ee7b87`). The manuscript’s own pin is `4b874cea0`. Its bibliography item for the source report is this report itself and is replaced by internal references; its items for Eriksson–Martin and Enouen are Part I’s [7, 8]; Gessel [10] and Taylor [11] were added to the bibliography.

Where the merge had to choose.

- Formulas already displayed in Part I (the third-order logarithmic expansion, $\log p_{n,k}$, $p_{n,k}$, the $k = 2, \dots, 5$ coefficients, the local masses, $\log B_{n,k}$ with A_1, A_2, A_3 , the ratio) are cited, not reprinted; the manuscript’s derivations of them are kept, and its contraction proof of (20), its ratio argument and its local computation appear as marked second proofs. The new polynomials q_1, q_2, q_3 are printed in (70).
- Symbols were renamed as listed in Section II.1; no formula changed its content.
- The manuscript’s Section 7 described its archive; it is rewritten to name the shipped files.
- Nothing of the manuscript’s mathematics, scope statements or disclaimers was removed.

Checks by the intake. The intake reran both checks on a copy (SymPy 1.14.0 under Python 3.13.5 and 3.14.4 on Windows); their outputs equal the recorded ones apart from line endings. Independently of the delivered scripts it confirmed $c_{n,M} \leq \lambda^M/M!$ in exact arithmetic for $k = 2, \dots, 7$, $n \leq 40$; the inequality of Theorem II.4.2 at sixteen points of $|v| = 2$ for $k = 2, 4$, $n = 5, 20$, $J = 0, 1, 2$; that $\deg_v q_s = s + 1$ exactly for $s = 1, \dots, 4$ and $k = 3, 4$ (so the degree bound of Theorem II.6.1 is attained); that q_3 equals the n^{-3} term of (40); and that the third-order residual of $\log \Phi_{n,k}(v)$ times N^4 is nearly constant at $n = 60, 120, 240$ for $k = 2, \dots, 5$ and several v . A naive principal logarithm failed at $k = 5$, $v = i$, which shows that the branch convention (67) is needed. These checks do not replace an independent review of the proofs.

II.A Direct verification of the weighted gamma tail

This appendix is unnecessary for Theorem II.4.2, but verifies the tail assertion in the original approach of Theorem 6.2. Let $T_N \sim \text{Gamma}(N + 1, 1)$, with density $t^N e^{-t}/N!$. The exact representation (27) is

$$\Phi_{n,k}(v) = \mathbb{E} R_k(v/T_N)^n.$$

For $E_N = \{t : |t - N| > N^{2/3}\}$, positivity and finite expansion give

$$\mathbb{E}[|R_k(v/T_N)|^n \mathbf{1}_{E_N}(T_N)] \leq \sum_{M=0}^d c_{n,M} V^M \mathbb{P}\{\text{Gamma}(N - M + 1, 1) \in E_N\}, \quad |v| \leq V. \quad (72)$$

Multiplying the original density by t^{-M} changes its shape to $N - M + 1$, with normalizing factor $(N - M)!/N!$, which proves (72). All shapes are positive.

Split at $M_N = \lfloor N^{1/3} \rfloor$. For $M \leq M_N$, the new mean $q = N - M + 1$ is within $M_N + 1$ of N . For all large N , membership in E_N implies deviation from q by more than $N^{2/3}/2$. The gamma moment-generating function $(1 - s)^{-q}$ and Chernoff optimization give

$$\mathbb{P}(T - q \geq a) \leq e^{-a^2/[2(q+a)]}, \quad \mathbb{P}(T - q \leq -a) \leq e^{-a^2/(2q)} \quad (0 < a < q).$$

For the upper bound optimize at $s = a/(q + a)$; for the lower, at $s = -a/(q - a)$. The resulting rate functions dominate the stated quadratic expressions by integrating their derivatives. Thus the $M \leq M_N$ contribution in (72) is at most $2e^{\lambda V - cN^{1/3}}$ for an absolute $c > 0$ and sufficiently large N .

The other terms total at most $\sum_{M > M_N} (\lambda V)^M / M!$. For fixed $a = \lambda V > 0$, once $M_N + 2 \geq 2a$ this is bounded by

$$\frac{2a^{M_N+1}}{(M_N + 1)!} \leq 2 \left(\frac{ea}{M_N + 1} \right)^{M_N+1} = \exp\{-\Omega(N^{1/3} \log N)\}.$$

For $V = 0$ it vanishes. The entire weighted complement is therefore smaller than every power of N^{-1} , uniformly on the disk. The same factorial bound for degrees $M \geq \varepsilon N$, fixed $\varepsilon > 0$, gives $\exp\{-\varepsilon N \log N + O(N)\}$ (or zero if beyond support). No uncontrolled prefactor C^n is introduced.

Part III

A second uniform tail repair and the fourth-order terms

Editorial note (5 October 2026, batch 101). Part III is bundle Report 98 of batch 101 of ProveIt’s incoming-reports intake, *A uniform tail repair for balanced Smirnov asymptotics*, dated 2 October 2026, with an empty author line. It arrived in the archive `A330266_Uniform_Tail_Repair.zip`; AI-assisted, unrefereed, not formally verified. Like Part II, it was written from this report as it stood at commit `4b874cea0`, when the report held Part I only, and it repairs the same step: the tail estimate of Theorem 6.2 that Remark 6.3 records as a sketch. Neither manuscript saw the other. Part III is therefore a second, independent answer to the assignment that Part II already answers, and most of its mathematics re-proves Part II theorem by theorem; Section III.2 maps every item of the manuscript to its place in Parts I and II. Printed in full are the items that are new relative to Parts I–II: the fourth-order terms (Section III.7), the factorial-moment bound for unequal rank sizes (Proposition III.4.2), a constant sharper than that of Theorem II.4.2 by the factor $2^{-(J+1)}$ (Remark III.5.1), and an open question on divergence and optimal truncation (Section III.9). The manuscript’s different proofs are kept as marked second routes; its proofs that agree with Part II step by step are cited. The write renamed the manuscript’s symbols to Part II’s (Section III.1), gave its labels the sub-prefix `bsw:ut:`, added the statements and notes marked [write], and collected the provenance in Section III.11.

Summary (the manuscript’s opening paragraphs). The analytic gap identified in Theorem 6.2 and Remark 6.3 is closed by the coefficientwise inequality $\mathbb{E}(X_{n,k})_{\underline{M}} \leq (k-1)^M$ for all $n \geq 1$ and $M \geq 0$. It yields an explicit uniform remainder for the differential-operator expansion; no gamma-tail argument and no path-forest theorem is needed. The note supplies the missing proof, proves the logarithmic degree bound needed for the cumulant claims, checks the stated correction terms, and gives the next probability coefficient. The Poisson limit and the leading $e^{-(k-1)}$ law were already proved in Part I; they are not new results of this repair. Throughout the asymptotic statements $k \geq 2$ is fixed and $n \rightarrow \infty$. The note is an AI-assisted, unrefereed mathematical proof and reproducibility package, not a formalization or a priority claim. [Write: when the manuscript was written, the gap had already been closed by Part II, which arrived in commit `096ee7b87` and was written into this report on 2 October 2026 (`1a372a1ca`). Part III’s repair is the second; Remark II.6.3 records the first.]

III.1 Notation of Part III

[Added at the write, 5 October 2026.] Part III uses the notation of Parts I and II. The manuscript agrees with them on $N = kn$, $\lambda = k - 1$, R_k , $a_{k,j}$, $r_{n,M}$, $c_{n,M}$, $H_r(M)$, $D = v d/dv$, $\Phi_{n,k}$, $p_{n,k}$, $B_{n,k}$, $F_N(v) = R_k(v/N)^n$, N_0 , $g(x)$, $(x)_{\underline{M}}$ and the Touchard polynomials; it writes m for the moment index that Part II calls M . Its other symbols collide with Parts I–II, or with each other, and are printed as follows.

Part III	Report 98	meaning, and the reading to avoid
$q_s(v)$	$Q_j(v)$	the coefficient of N^{-s} in $\log \Phi_{n,k}$, as in (69); <i>not</i> Part II's product $Q_M(x) = \prod_{j < M} (1 - jx)^{-1}$
$\mathcal{P}_s(v)$	$P_j(v)$	the coefficient of N^{-s} in $e^{-\lambda v} \Phi_{n,k}$, as in (66); <i>not</i> the Laguerre kernels $P_k(t)$ of (13)
$\mathcal{P}_4(-1)$	$C_4(k)$	the fourth probability coefficient; <i>not</i> the reciprocal polynomial $C_4(z)$ of (16) evaluated at $z = k$
J	q	the truncation order (q now names the log coefficients)
$\mathcal{C}_{k,V,J}$	$C_{q,k,V}$	the remainder constant of Theorem II.4.2; <i>not</i> $C_k(z)$
$f_s(v)$	$A_j(v)$	the expansion coefficients of $e^{-\lambda v} F_N(v)$, as in (64); <i>not</i> the Stirling constants A_1, A_2, A_3 of (50), which the manuscript also uses
$Q_M(1/N)$	$A_{N,m}$	$N^M / (N)_{\underline{M}}$
b_j, γ_j	$\ell_j; b_j$	the manuscript's $\ell_j = [z^j] \log R_k$ is Part II's b_j (with $b_1 = k\lambda$); the manuscript's $b_j = \ell_j/k$ (with $b_1 = \lambda$) is $1/k$ times Part II's; its $b_{j+1}v^{j+1}$ is Part II's γ_j
w	h	the expansion variable $1/N$; h_s stays the complete homogeneous symmetric polynomial
$\sigma_s(D)/s, \mathcal{A}_w(D)$	$L_s(D), \mathcal{L}(h, D)$	the power-sum generator (68); L stays $\log Y + \lambda$
$\Sigma(t, w, v)$	$U(t, h, v)$	the formal logarithm along the operator flow
$\mathcal{B}_a$	B_d	$e^{-\Sigma} D^a e^{\Sigma} 1$; <i>not</i> the count $B_{n,k}$
$\psi(x)$	$f(x)$	the inverse carrier $x(\log x - 1)$; <i>not</i> f_s
M_0	L	the integer cut of the tail bound (74); <i>not</i> $L = \log Y + \lambda$, and <i>not</i> the cut M_N of Appendix II.A
$k_{\max}$	K	the largest rank size in Section III.4; <i>not</i> the graphs $K_{k,\dots,k}$ or the cliques K_k

Normalization: like Part II, the manuscript expands in powers of $N^{-1} = (kn)^{-1}$; a coefficient of N^{-s} is k^{-s} times the corresponding coefficient of n^{-s} . For $k = 4$, the manuscript's $\mathcal{P}_4(-1) = 332775/5760$ becomes $332775/(5760 \cdot 4^4) = 7395/32768$.

III.2 Scope, and a map to Parts I and II

[Added at the write.] Every item of the manuscript is listed below with its place in this report. “Same proof” means that the two arguments agree step by step (the same inequality $h_{a+b} \leq h_a h_b$, the same formal flow); such items are cited, *not* printed again.

Report 98	content	in this report
Section 1	$\Phi_{n,k}(v) = \sum_M c_{n,M} v^M$ by marked directed path forests	(20); third proof printed in Section III.3
Lemma 1, first part	$\mathbb{E}(X)_{\underline{M}} \leq \lambda^M$	Theorem II.3.1; second route printed in Section III.4
Lemma 1, second part	the tail bound	new, elementary: (74)
$H_s(m), H_1-H_3$	complete homogeneous polynomials	(28)–(32); H_4 in Section III.7
Lemma 2	positive scalar remainder	Lemma II.4.1 at $x = 1/N$; same proof
Theorem 3	uniform operator remainder	Theorem II.4.2; same proof; sharper constant in Remark III.5.1

Report 98	content	in this report
Theorem 4, first part	collected expansion, $\deg \mathcal{P}_s \leq 2s$	Corollary II.5.1; same proof; degree bound in Remark III.6.1
Theorem 4, second part	branch, $\deg q_s \leq s + 1$, cumulant scale	(67), Theorem II.6.1; same proof
Section 5, Q_1 – Q_3	q_1, q_2, q_3	(70), which is Theorem 8.1
Section 5, p to N^{-3}	no-adjacency probability	(42)
Section 5, C_4 , $Q_4(-1)$, $k = 2, \dots, 6$	fourth-order terms	new: Theorem III.7.1, Corollary III.7.2
Section 5, local law	first local correction	Corollary 8.3 and Part II's second derivation; same computation; all orders in Remark III.7.5
Section 6, items 1–4	what the repair restores; the ratio	Remark II.6.3; (52) with Part II's second proof; same argument
Section 6, inverse	Lambert- W inverse	(71); same proof; smooth carrier in Remark III.8.1
Section 7, question 1	growing multiplicity	Part I question 1; question 1 below
Section 7, question 2	beyond all algebraic orders	new: question 2 below
Section 7, question 3	unequal rank sizes	Propositions III.3.1–III.4.2; Part I question 4; question 3 below
Section 8	sources and checks	Sections III.10 and III.11

The manuscript's scope is Part II's: $k \geq 2$ is fixed; asymptotic statements hold for a fixed truncation order, on compact v -disks, with the factorial-cumulant order r and the local index j fixed. Its factorial-moment bounds hold for all n, k and M . It does not newly prove the leading Poisson law or the $e^{-(k-1)}$ limit, which are Part I's.

III.3 The exact identity by marked directed path forests

Part I proves (20) by a weighted run transform and the Laguerre integral, and Part II adds a contraction proof on multiset words. The manuscript counts labelled cards rank by rank instead.

Third proof of (20) (Report 98). Mark M equal-rank adjacencies of a permutation. Within each rank they form a directed union of vertex-disjoint paths on its k labelled cards: a card has at most one marked successor and at most one marked predecessor, and the marked edges follow the left-to-right order. If the rank has j marked edges, the forest has $k - j$ path components, isolated cards included. The directed path forests with j edges on k labelled cards number

$$\frac{k!}{(k-j)!} \binom{k-1}{j} = a_{k,j} :$$

arrange the k labels in a row, keep j of the $k - 1$ intervening bonds, and divide by the $(k - j)!$ orders of the resulting components. Choose such a forest for every rank, with M edges in all, and contract each path to a block. The $N - M$ blocks can be arranged in $(N - M)!$ orders, and each arrangement is a permutation in which the marked edges are equal-rank adjacencies. Further, unmarked, equal-rank adjacencies are allowed. This is a bijection onto pairs of a permutation and an M -element subset of its equal-rank adjacencies. Hence $\sum_{\pi} \binom{X(\pi)}{M} = r_{n,M}(N - M)!$, which is (20) after division by $N!$. In particular the support degree is $n(k - 1) = N - n < N$. $\square$

[Write: $a_{k,j} = \frac{k!}{(k-j)!} \binom{k-1}{j}$ is the Lah number $L(k, k-j)$, the number of partitions of k labelled cards into $k-j$ nonempty linearly ordered blocks; Section 11 of Part I identifies the same numbers as rook numbers of a $k \times (k-1)$ board.]

The count is local to each rank, so it applies verbatim to ranks of different sizes. The manuscript states this in its third further question; the write states and proves it.

Proposition III.3.1 (Unequal rank sizes: the exact identity; Report 98, proof written out at the write). *Let $\mathbf{k} = (k_1, \dots, k_n)$ with $n \geq 1$ and integers $k_i \geq 1$, put $N = k_1 + \dots + k_n$, and let $X_{\mathbf{k}}$ count the equal-rank adjacencies in a uniform permutation of N distinct cards, k_i of them of rank i . Define $a_{k,j}$ and R_k by (14)–(15) also for $k = 1$, so that $R_1 = 1$, and put $r_{\mathbf{k},M} = [z^M] \prod_{i=1}^n R_{k_i}(z)$. Then, for $0 \leq M \leq N - n$,*

$$\mathbb{E} \binom{X_{\mathbf{k}}}{M} = \frac{r_{\mathbf{k},M}}{(N)_{\underline{M}}}, \quad (73)$$

and for $M > N - n$ the left side vanishes and $r_{\mathbf{k},M} = 0$ (the quotient is not used there). In particular $\mathbb{E} X_{\mathbf{k}} = \sum_i k_i(k_i - 1)/N$; when every $k_i = 1$ both sides are zero.

Proof. Rank i carries at most $k_i - 1$ marked edges. Repeat the third proof above with a_{k_i, j_i} directed path forests of j_i edges for rank i (for $k_i = 1$ only $j_i = 0$ occurs, with one forest). Choosing forests with $j_1 + \dots + j_n = M$ edges in all, contracting and arranging the $N - M$ blocks gives $\sum_{\pi} \binom{X_{\mathbf{k}}(\pi)}{M} = (N - M)! \sum_{j_1 + \dots + j_n = M} \prod_i a_{k_i, j_i} = (N - M)! r_{\mathbf{k},M}$, the bijection being the same. Divide by $N!$; the quotient is defined because $M \leq N - n < N$. For $M > N - n = \sum_i (k_i - 1)$ no choice of forests exists, and no permutation has more than $N - n$ equal-rank adjacencies. If some $k_i \geq 2$, then $N - n \geq 1$, and the case $M = 1$ gives $\mathbb{E} X_{\mathbf{k}} = r_{\mathbf{k},1}/N$ with $r_{\mathbf{k},1} = \sum_i a_{k_i,1} = \sum_i k_i(k_i - 1)$. If every $k_i = 1$, then $N - n = 0$, so $X_{\mathbf{k}} = 0$, and $\sum_i k_i(k_i - 1) = 0$ as well. $\square$

[Dated note, 5 October 2026: the statement first read “Let $\mathbf{k} = (k_1, \dots, k_n)$ with integers $k_i \geq 1$ ” and ended “and both sides vanish for $M > N - n$. In particular $\mathbb{E} X_{\mathbf{k}} = \sum_i k_i(k_i - 1)/N$ ”, and the proof ended “For $M = 1$, $r_{\mathbf{k},1} = \sum_i a_{k_i,1} = \sum_i k_i(k_i - 1)$ ”. An independent check made by the intake after the write found the proposition valid and its wording imprecise in three places: for $M > N$ the right side is $0/0$, not zero; the mean was read off the case $M = 1$, which lies in the range only when $N - n \geq 1$, so compositions with every $k_i = 1$ (mean zero) were not covered by the cited case; and $n \geq 1$ was implicit. The present wording is the check’s; see the end of Section III.11.]

III.4 The factorial-moment majorant

Theorem II.3.1 bounds $c_{n,M}$ through the majorant $(1 + kz)^{k-1}$ and the inequality $(d - j)/(N - j) \leq d/N$. The manuscript’s Lemma 1 uses the majorant $(1 + \lambda z)^k$, which needs no second step.

Second proof of Theorem II.3.1 (Report 98). For $0 \leq j \leq k - 1$ the j factors of $(k - 1)_{\underline{j}}$ lie in $[0, \lambda]$, so $a_{k,j} = \binom{k}{j} (k - 1)_{\underline{j}} \leq \binom{k}{j} \lambda^j$. Hence

$$R_k(z) \preceq (1 + \lambda z)^k, \quad R_k(z)^n \preceq (1 + \lambda z)^N,$$

since products of polynomials with nonnegative coefficients preserve $\preceq$. For $M \leq d = n(k - 1)$, dividing $r_{n,M} \leq \binom{N}{M} \lambda^M$ by $(N)_{\underline{M}} = M! \binom{N}{M}$ gives $c_{n,M} \leq \lambda^M / M!$; beyond the support $c_{n,M} = 0$. $\square$

Corollary III.4.1 (Tail bound; Report 98). *For $V \geq 0$ and every integer $M_0 \geq 0$,*

$$\sup_{|v| \leq V} \left| \sum_{M \geq M_0} c_{n,M} v^M \right| \leq \sum_{M \geq M_0} \frac{(\lambda V)^M}{M!} \leq e^{\lambda V} \frac{(\lambda V)^{M_0}}{M_0!}. \quad (74)$$

Proof. Sum the bound of Theorem II.3.1 and use $(M_0 + j)! \geq M_0! j!$. $\square$

The manuscript notes that this majorant accounts for the growing number of rank cliques directly, and does not invoke the fixed-path-family hypotheses of the path-forest report `a189281-path-forest-expansions`. The inequality holds for every k ; only the asymptotic constants below depend on the fixed k . The same majorant works rank by rank.

Proposition III.4.2 (Unequal rank sizes: factorial-moment bound; Report 98). *In the setting of Proposition III.3.1, let $k_{\max} = \max_i k_i$. For every integer $M \geq 0$,*

$$0 \leq \mathbb{E} \binom{X_{\mathbf{k}}}{M} \leq \frac{(k_{\max} - 1)^M}{M!}, \quad \mathbb{E}(X_{\mathbf{k}})_{\underline{M}} \leq (k_{\max} - 1)^M, \quad (75)$$

and $|\mathbb{E}(1 + v)^{X_{\mathbf{k}}}| \leq e^{(k_{\max} - 1)|v|}$ for complex v .

Proof. For $0 \leq j \leq k_i - 1$ the j factors of $(k_i - 1)_j$ lie in $[0, k_{\max} - 1]$, so $a_{k_i,j} \leq \binom{k_i}{j} (k_{\max} - 1)^j$ and $R_{k_i}(z) \preceq (1 + (k_{\max} - 1)z)^{k_i}$. The product over i gives $\prod_i R_{k_i}(z) \preceq (1 + (k_{\max} - 1)z)^N$, so $r_{\mathbf{k},M} \leq \binom{N}{M} (k_{\max} - 1)^M$. Divide by $(N)_{\underline{M}} = M! \binom{N}{M}$ for $M \leq N - n$, using (73); for $M > N - n$ the left side is zero. The last assertion follows by summing absolute values. (If $k_{\max} = 1$ then $X_{\mathbf{k}} = 0$, and the bound reads 0^M with $0^0 = 1$.) $\square$

The manuscript does not claim that this bound alone gives a limit law or an expansion for varying compositions. The exact identity does give the limit law when the multiplicities stay bounded.

Proposition III.4.3 (Bounded multiplicities: Poisson limit; write). *[Write, 5 October 2026.] Fix $K \geq 1$. For each n let $\mathbf{k} = (k_1, \dots, k_n)$ with $1 \leq k_i \leq K$, let $N = \sum_i k_i$, and put $\mu_n = \mathbb{E} X_{\mathbf{k}} = \sum_i k_i(k_i - 1)/N$. For every fixed $M \geq 0$,*

$$\mathbb{E}(X_{\mathbf{k}})_{\underline{M}} = \mu_n^M + O_{K,M}(n^{-1}).$$

Consequently, if $\mu_n \rightarrow \mu$ as $n \rightarrow \infty$, then $X_{\mathbf{k}} \rightarrow \text{Poisson}(\mu)$ in distribution, and $\mathbb{P}(X_{\mathbf{k}} = 0) \rightarrow e^{-\mu}$.

Proof. Put $x_i = a_{k_i,1} = k_i(k_i - 1) \leq K^2$, so $\sum_i x_i = \mu_n N$. In $r_{\mathbf{k},M} = \sum_{j_1 + \dots + j_n = M} \prod_i a_{k_i,j_i}$, the terms with every $j_i \leq 1$ sum to the elementary symmetric polynomial $e_M(x_1, \dots, x_n)$. In every other term some $j_i \geq 2$, so at most $M - 1$ indices have $j_i \geq 1$. There are at most Mn^{M-1} such index sets and at most 2^{M-1} ways to distribute M among a given set, and each product is at most $2^{KM} K^M$, because $a_{k,j} \leq \binom{k}{j} (k - 1)^j \leq 2^K K^j$ for $k \leq K$, at most M factors differ from $a_{k_i,0} = 1$, and the j_i sum to M . Hence these terms sum to $O_{K,M}(n^{M-1})$. Next, $(\sum_i x_i)^M$ is the sum of $x_{i_1} \cdots x_{i_M}$ over all M -tuples of indices; the tuples with distinct entries contribute $M! e_M(x)$, and the at most $\binom{M}{2} n^{M-1}$ tuples with a repeated entry contribute at most $\binom{M}{2} n^{M-1} K^{2M}$. Therefore

$$r_{\mathbf{k},M} = \frac{(\mu_n N)^M}{M!} + O_{K,M}(n^{M-1}).$$

If $N < 2M$, then $n \leq N < 2M$ is bounded, and both $\mathbb{E}(X_{\mathbf{k}})_{\underline{M}}$ and μ_n^M lie in $[0, (K-1)^M]$ by Proposition III.4.2, so the claim holds trivially. Let $N \geq 2M$, so that $(N)_{\underline{M}} \geq (N/2)^M$ and $N^M/(N)_{\underline{M}} = 1 + O_M(N^{-1})$; recall $n \leq N$ and $\mu_n \leq K-1$. If $M \leq N-n$, Proposition III.3.1 gives

$$\mathbb{E}(X_{\mathbf{k}})_{\underline{M}} = \frac{M! r_{\mathbf{k},M}}{(N)_{\underline{M}}} = \mu_n^M \frac{N^M}{(N)_{\underline{M}}} + O_{K,M}\left(\frac{n^{M-1}}{N^M}\right) = \mu_n^M + O_{K,M}(n^{-1}).$$

If $M > N-n$, then $\mathbb{E}(X_{\mathbf{k}})_{\underline{M}} = 0$ and $r_{\mathbf{k},M} = 0$, and the display above gives $(\mu_n N)^M = O_{K,M}(n^{M-1})$, that is $\mu_n^M = O_{K,M}(n^{M-1}/N^M) = O_{K,M}(n^{-1})$. If $\mu_n \rightarrow \mu$, every factorial moment converges to μ^M , the M th factorial moment of $\text{Poisson}(\mu)$ (for $\mu = 0$, of the point mass at 0). These laws are determined by their moments, so the method of moments gives the convergence in distribution, exactly as in the proof of Theorem 5.1; the mass at 0 converges because $\pm \frac{1}{2}$ carry no limiting mass. $\square$

Remark III.4.4 (What this does for Part I's question 4). [Write.] Research question 4 of Section 14 asks for natural conditions under which the adjacency count of unequal ranks is asymptotically Poisson with mean $\sum_i k_i(k_i - 1)/N$. That mean is exact (Proposition III.3.1), and bounded multiplicities with a convergent mean suffice (Proposition III.4.3). The proof is the standard method of moments; no priority is claimed. Unbounded multiplicities and the correction terms in symmetric sums of the k_i remain open (question 3 of Section III.9).

III.5 The operator remainder and a sharper constant

The manuscript's Lemma 2 is Lemma II.4.1 at $x = 1/N$, where $Q_M(1/N) = N^M/(N)_{\underline{M}}$, proved the same way: splitting a weakly increasing index string after a entries injects the terms of h_{a+b} into those of $h_a h_b$. Its Theorem 3 is Theorem II.4.2, again with the same proof: multiplying the scalar remainder by $r_{n,M}|v|^M/N^M$ turns the factor $Q_M(1/N)$ back into $c_{n,M}$, and Theorem II.3.1 then controls the whole support at once. In the manuscript's words, the estimate holds for the full degree range, and no growing cutoff, exchange of an uncontrolled infinite sum, gamma concentration estimate, or cancellation of half-integer powers is involved. The two differ only in the constant.

Remark III.5.1 (A sharper constant; Report 98). With $a = \lambda V$, the constant of (61) satisfies

$$C_{k,V,J} = \sum_{M \geq 0} H_{J+1}(M) \frac{a^M}{M!} \leq 2^{-(J+1)} e^a \mathcal{T}_{2J+2}(a). \quad (76)$$

Indeed $h_{a+b} \leq h_a h_b$, iterated, gives $H_{J+1}(M) \leq H_1(M)^{J+1} = (M(M-1)/2)^{J+1} \leq M^{2J+2}/2^{J+1}$, and $\sum_M M^{2J+2} a^M/M! = e^a \mathcal{T}_{2J+2}(a)$. Part II's (58) records the middle bound and then discards the factor 2^{-s} , which costs the factor 2^{J+1} in (61). The refinement is elementary and changes no asymptotic statement.

Remark III.5.2 (Unequal rank sizes; write). The proof of Theorem II.4.2 uses only the identity $c_{n,M} = r_{n,M}/(N)_{\underline{M}}$ on a support below N and the bound $c_{n,M} \leq \lambda^M/M!$. By Propositions III.3.1 and III.4.2 both hold for unequal rank sizes with λ replaced by $k_{\max} - 1$. Hence, with $F_{\mathbf{k}}(v) = \prod_i R_{k_i}(v/N)$ and $a = (k_{\max} - 1)V$, the same argument gives, for every $J \geq 0$, $V \geq 0$ and composition $\mathbf{k}$,

$$\sup_{|v| \leq V} \left| \mathbb{E}(1+v)^{X_{\mathbf{k}}} - \sum_{r=0}^J N^{-r} H_r(D) F_{\mathbf{k}}(v) \right| \leq \frac{1}{N^{J+1}} \sum_{M \geq 0} H_{J+1}(M) \frac{a^M}{M!},$$

and the right side is at most $2^{-(J+1)} e^a \mathcal{T}_{2J+2}(a)/N^{J+1}$ by Remark III.5.1. Turning this operator rule into a collected expansion is part of question 3.

III.6 All orders and the logarithmic degree

The manuscript's Theorem 4 is Corollary II.5.1 together with Theorem II.6.1, with the same proofs. In Part II's notation: its $A_j(v)$ are the $f_j(v)$ of (64), its P_j are the $\mathcal{P}_j$ of (66), its normalized logarithm is the branch (67), its flow $U = \log(\exp(t\mathcal{L}(h, D))e^f)$ is $\Sigma(t, w, v)$ with the generator $\mathcal{A}_w(D)$ of (68), its recursion $B_{d+1} = DB_d + (DU)B_d$ is that of $\mathcal{B}_a$, and its induction on j is the induction on s in the proof of Theorem II.6.1. Two small statements of the manuscript are not in Part II.

Remark III.6.1 (Degree of $\mathcal{P}_s$; Report 98). $\deg_v \mathcal{P}_s \leq 2s$. Indeed H_r has degree at most $2r$, the conjugated operator $e^{-\lambda v} D e^{\lambda v} = D + \lambda v$ raises the degree of a polynomial by at most one, so $e^{-\lambda v} H_r(D) e^{\lambda v} = H_r(D + \lambda v)$ raises it by at most $2r$, and $\deg f_{s-r} \leq 2(s-r)$; apply this to each term of (66). The bound is attained for $s \leq 4$ (Theorem III.7.1).

Remark III.6.2 (Vanishing low coefficients; Report 98). Every q_s , $s \geq 1$, has zero constant and linear coefficients: the branch (67) vanishes at $v = 0$, and $\kappa_1^{(F)} = \mathbb{E}X = \lambda$ exactly. As in Part II, the logarithm is this normalized analytic branch, not the principal logarithm on all compact subsets of $\mathbb{C}$. The cumulant scale holds for each fixed r , with constants depending on r ; nothing uniform in all cumulant orders is claimed.

III.7 The fourth-order terms

The expansions of Parts I and II stop at N^{-3} . The manuscript computes the next coefficient for every k . Part I's $H_0, \dots, H_3$ of (29)–(32) are followed by

$$H_4(M) = \frac{M(M-1)(M+1)(M+2)(M+3)(15M^3 - 15M^2 - 10M + 8)}{5760}. \quad (77)$$

Theorem III.7.1 (Fourth-order terms; Report 98). *Fix $k \geq 2$ and put $\lambda = k - 1$. The coefficients of N^{-4} in (69) and (65) are*

$$q_4(v) = \frac{\lambda^2 v^2}{60} \{12(k-2)(k^2 - 12k + 17)v^3 - 15(7k^2 - 38k + 43)v^2 + 140(k-2)v - 30\}, \quad (78)$$

$$\begin{aligned} \mathcal{P}_4(v) = \frac{\lambda^2 v^2}{5760} \{ & 15\lambda^6 v^6 + 240(k-2)\lambda^4 v^5 + 40\lambda^2(17k^2 - 122k + 149)v^4 \\ & - 384(k-2)(7k^2 + 16k - 41)v^3 - 720(11k^2 - 70k + 83)v^2 + 13440(k-2)v - 2880 \}. \end{aligned} \quad (79)$$

Consequently, uniformly for fixed k ,

$$p_{n,k} = e^{-\lambda} \left[1 - \frac{\lambda^2}{2N} + \frac{\lambda^2(3k^2 - 14k + 7)}{24N^2} - \frac{\lambda^4(k^2 - 10k + 17)}{48N^3} + \frac{\mathcal{P}_4(-1)}{N^4} + O_k(N^{-5}) \right], \quad (80)$$

$$\mathcal{P}_4(-1) = \frac{\lambda^2}{5760} (15k^6 - 330k^5 + 2345k^4 - 7212k^3 + 13313k^2 - 10122k + 2183),$$

and the new logarithmic coefficient, the coefficient of N^{-4} in $\log p_{n,k}$, is

$$q_4(-1) = -\frac{\lambda^2(12k^3 - 63k^2 + 62k - 13)}{60}. \quad (81)$$

Proof. Corollary II.5.1 and Theorem II.6.1 with $J = 4$ give the form of both expansions and the remainders $O_k(N^{-5})$, uniformly on $|v| \leq 1$, hence at $v = -1$. The coefficients follow from (66), with $f_0, \dots, f_4$ of (64) (they need $b_1, \dots, b_5$) and $H_0, \dots, H_4$, by finite algebra, and q_4 from the logarithmic series $\sum_{s \geq 1} q_s w^s = \log \sum_{s \geq 0} \mathcal{P}_s w^s$ to order w^4 . The terms through N^{-3} are those of (42) and (70). This algebra is mechanical but long; the manuscript performs it by computer algebra for symbolic k , and so does this proof: it is an exact computer-algebra identity, not a hand derivation (Section III.10 names the programs and the independent second computation). $\square$

[Write: the shipped polynomials pass three consistency checks made at the write: $\log \sum_{s \leq 4} \mathcal{P}_s w^s$ reproduces $q_1, \dots, q_4$ to order w^4 ; (79) at $v = -1$ equals the displayed $\mathcal{P}_4(-1)$; and (78) at $v = -1$ equals (81). The v^5 coefficient of q_4 is $\lambda^2(k-2)(k^2-12k+17)/5$, which is nonzero for every integer $k \geq 3$, since $k^2-12k+17$ has the irrational roots $6 \pm \sqrt{19}$; so $\deg q_4 = 5$ for $k \geq 3$, the bound of Theorem II.6.1, and $\deg q_4 = 4$ for $k = 2$. Likewise $\deg \mathcal{P}_4 = 8$, the bound of Remark III.6.1.]

These are exact algebraic consequences of the proved all-orders expansion; in the manuscript's words, the package does not rely on fitting numerical residuals.

Corollary III.7.2 (The OEIS rows; Report 98, last two columns write). *In powers of n^{-1} , the coefficient of n^{-4} in $e^{k-1}p_{n,k}$ is $\mathcal{P}_4(-1)/k^4$, and that in $\log p_{n,k}$ is $q_4(-1)/k^4$. For A330266 ($k = 4$),*

$$p_{n,4} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + \frac{7395}{32768n^4} + O(n^{-5}) \right), \quad (82)$$

and $\log p_{n,4} = -3 - \frac{9}{8n} - \frac{21}{32n^2} - \frac{81}{256n^3} + \frac{3}{1024n^4} + O(n^{-5})$. For $k = 2, \dots, 6$:

k	$[n^{-4}] e^{k-1}p_{n,k}$	$[n^{-4}] \log p_{n,k}$	$A_4 = q_4(-1)$
2	361/6144	3/64	3/4
3	110/729	14/243	14/3
4	7395/32768	3/1024	3/4
5	2152/9375	-296/3125	-296/5
6	179035/1492992	-3415/15552	-3415/12

The second column is the manuscript's; the third and fourth were computed at the write from (81). The rows $k = 2, 3, 4$ are A007060, A193624 and A330266 [2, 3, 1].

Remark III.7.3 (The count expansion to N^{-4} ; write). The count expansion (49) extends to

$$\log B_{n,k} = N(\log N - 1) + \frac{1}{2} \log(2\pi N) - \lambda + \sum_{s=1}^4 \frac{A_s}{N^s} + O_k(N^{-5}), \quad A_4 = q_4(-1). \quad (83)$$

Indeed Stirling's series $\log N! = N(\log N - 1) + \frac{1}{2} \log(2\pi N) + \frac{1}{12N} - \frac{1}{360N^3} + O(N^{-5})$ has no even inverse powers, and $\log B_{n,k} = \log N! + \log p_{n,k}$ with $\log p_{n,k} = -\lambda + \sum_{s \leq 4} q_s(-1)N^{-s} + O_k(N^{-5})$ by Theorem II.6.1. The same bookkeeping reproduces (50): $A_1 = \frac{1}{12} + q_1(-1)$, $A_2 = q_2(-1)$, $A_3 = -\frac{1}{360} + q_3(-1)$.

Remark III.7.4 (The standard deck; write). [Computed from the exact count.] At $n = 13$, $k = 4$, adding the n^{-4} term of (82) to the third-order approximation of Section 9.1 gives 0.0454762434419780, an error of -3.89×10^{-8} against the exact 0.0454762823310943 of (48); through n^{-3} the error is -4.32×10^{-7} . The exponential of the logarithmic form through n^{-4} gives 0.0454762297566350, error -5.26×10^{-8} (through n^{-3} : -5.72×10^{-8}). These are diagnostics, not part of any proof.

The manuscript's first local correction is (43) of Corollary 8.3, derived as in Part II's second derivation: put $v = u - 1$ and use Cauchy's coefficient formula on a fixed u -circle. It adds that every further fixed- j correction can be extracted in the same way. The write records why.

Remark III.7.5 (Local probabilities to every order; write). For fixed integers $j \geq 0$ and $J \geq 0$,

$$\mathbb{P}(X_{n,k} = j) = \sum_{s=0}^J N^{-s} [u^j] [e^{\lambda(u-1)} \mathcal{P}_s(u-1)] + O_{k,j,J}(N^{-J-1}).$$

Indeed $\Phi_{n,k}(u-1) = \mathbb{E}u^X = \sum_i \mathbb{P}(X = i)u^i$, and on $|u| = 1$ one has $|v| = |u-1| \leq 2$, so Corollary II.5.1 with $V = 2$ bounds the difference between $\Phi_{n,k}(u-1)$ and the truncated sum by $O_{k,J}(N^{-J-1})$ uniformly on the circle. The coefficient of u^j of a function analytic on $|u| \leq 1$ is bounded by its maximum on $|u| = 1$. Each coefficient on the right is a finite sum, since $\mathcal{P}_s$ is a polynomial: with $\mathcal{P}_s(u-1) = \sum_i \pi_{s,i}u^i$, it is $e^{-\lambda} \sum_{i \leq j} \pi_{s,i} \lambda^{j-i} / (j-i)!$. For $J = 1$ this is (43). The statement is for fixed j ; nothing uniform in a growing j is claimed.

III.8 Counts, ratios, the inverse, and what the repair restores

The manuscript's Section 6 lists the claims of Part I that its repair supports, for fixed k : Theorem 6.2, with the explicit estimate of Theorem II.4.2; the factorial-cumulant scale stated after Theorem 7.1; Theorem 8.1, Corollaries 8.2 and 8.3 and the specializations (44)–(47); the count and ratio expansions of Section 10; and the Lambert- W estimate with its proper scope. This is the list of Remark 6.3, and Remark II.6.3 draws the same conclusion from Part II. The manuscript re-derives the ratio expansion (52) by subtracting $\log p_{n,k}$ at consecutive integers, which is Part II's second proof, and the inverse (71) by the same Taylor argument: the mean-value theorem first gives $N_0 - N = O_k(1)$, then $\psi''(x) = 1/x$ and $g'(x) = O_k(1/x)$ give the error $O_k(1/(N_0 \log N_0))$. It notes that this is at least as strong as the $O_k(N_0^{-1})$ of (54) on that domain. It also states the repair's limits: no growing- k uniformity, no exponentially improved transseries or optimal truncation, no certified integer inverse, and no change in the status of Part I's exact formulas, exact first four cumulants, or leading Poisson law. One sentence goes beyond Part II: “for general Y , the same reasoning applies to the eventually increasing smooth carrier”.

Remark III.8.1 (Inverse of the smooth carrier; Report 98, proof written out at the write). Let $G(x) = \psi(x) + g(x)$ with ψ and g as in Section II.7.3. Since $G'(x) = \log x + O_k(1/x)$, there is x_k with G increasing on $[x_k, \infty)$. For $L = \log Y + \lambda \geq G(x_k)$ let $N_*(Y) \geq x_k$ be the unique solution of $G(N_*) = L$. Then, with N_0 of (53),

$$N_*(Y) = N_0 - \frac{g(N_0)}{\log N_0} + O_k\left(\frac{1}{N_0 \log N_0}\right) \quad (Y \rightarrow \infty).$$

[Write: the proof of (71) applies with its $O(N^{-4})$ term replaced by zero. From $\psi(N_*) + g(N_*) = \psi(N_0)$ and $g(x) = O_k(\log x)$, first $N_* \sim N_0$; the mean-value theorem with $\psi' = \log x$ then gives $N_0 - N_* = O_k(1)$; Taylor expansion with $\psi''(x) = 1/x$ and $g'(x) = O_k(1/x)$ gives $(N_* - N_0) \log N_0 + g(N_0) = O_k(1/N_0)$. Along the sequence, $G(N) = L + O_k(N^{-4})$ by (49), so $N_*(B_{n,k}) = N + O_k(N^{-4}/\log N)$.] As the manuscript says, an integer least-index claim requires rounding and bracketing information that is separate from this expansion; the remark certifies no integer threshold.

III.9 Further questions and research

The manuscript's three further questions, with what Parts I–III now prove. Its unproved method claim on further local corrections is proved in Remark III.7.5, so it is not listed.

1. **Growing multiplicity** (Report 98's first question; Part I's question 1). Determine ranges $k = k(n)$ for which the correction expansion is uniform with an error relative to $p_{n,k}$. The bound of Theorem II.3.1 holds for all k , but its constants grow with k : an absolute bound involving e^{k-1} does not by itself control a probability on the much smaller scale $e^{-(k-1)}$. Missing: quantitative control, in k , of the Taylor radius of $\log R_k$ and of the operator remainder.
2. **Beyond all algebraic orders** (Report 98's second question; new). Determine how q_s and $\mathcal{P}_s$ grow with s ; decide whether the fixed- k series $\sum_s q_s(v)N^{-s}$ converges or diverges; and, where relevant, find an optimal truncation rule with an exponentially improved remainder. The estimates of Parts II–III hold for each fixed order and give no such rule. A related gap, recorded by the manuscript's audit: the constant of Theorem II.4.2 is explicit, but the passage to the collected expansion of Corollary II.5.1 uses Cauchy estimates with unspecified constants, so no explicit finite- n constant for the collected expansion is known. Missing: everything beyond the five computed terms. [Write: Part V of the companion report `a189281-path-forest-expansions` takes up the divergent series of its own model, through a Borel completion along the positive ray with a nonzero flat remainder; nothing of it is transferred here.]
3. **Unequal rank sizes** (Report 98's third question; Part I's question 4). For rank sizes $k_1, \dots, k_n$ the exact identity, the bound $\mathbb{E}(X_{\mathbf{k}})_{\underline{M}} \leq (k_{\max} - 1)^M$, the Poisson limit for bounded multiplicities with a convergent mean, and the operator remainder are proved (Propositions III.3.1–III.4.3, Remark III.5.2). Open: expansions in the symmetric sums of the multiplicities, under explicit assumptions that give a stable limiting mean and uniform remainders; the manuscript names bounded maximum multiplicity as the natural first regime. [Write sketch: for $k_{\max} \leq K$ only finitely many R_k occur, all zero-free on a common disk $|z| \leq \rho_K$, so $\log F_{\mathbf{k}}(v) = \sum_{j \geq 1} m_j(\mathbf{k}) v^j N^{1-j}$ with the bounded symmetric sums $m_j(\mathbf{k}) = N^{-1} \sum_i [z^j] \log R_{k_i}(z)$, and the argument of Corollary II.5.1 should give an expansion whose coefficients are polynomials in v and the m_j . Missing: writing this out with uniform constants, and deciding how the m_j , which vary with n , enter the final statement.] Unbounded multiplicities are open.
4. **A certified integer inverse** (Part I's question 11). Remark III.8.1 adds the inverse of the smooth carrier for arbitrary Y ; the bracketing theorem for the least n with $B_{n,k} \geq Y$ remains open, as Part II states.

III.10 Sources and reproducibility

[Rewritten at the write to name the shipped files.] The manuscript's programs and records are shipped as follows.

shipped file	content
<code>code/03-tail-check_repair.py</code>	the main check
<code>code/03-tail-independent_check.py</code>	the independent check
<code>code/03-tail-run_checks.sh</code>	the delivery's driver script
<code>data/03-tail-symbolic_checks.txt</code>	recorded output of the main check
<code>data/03-tail-coefficients.json</code>	$\mathcal{P}_0, \dots, \mathcal{P}_4, q_1, \dots, q_4, H_0, \dots, H_4$
<code>data/03-tail-independent_checks.txt</code>	recorded output of the independent check
<code>data/03-tail-source_manifest.json</code>	source pin and Git blob identifiers
<code>03-tail-AUDIT.md</code>	the delivery's independent audit

The main check requires Python 3 and SymPy 1.14.0. It verifies the Git blob identifiers of four source files against the manifest; checks 14,850 exact factorial-coefficient inequalities $c_{n,M} \leq \lambda^M / M!$ for $2 \leq k \leq 10$, $1 \leq n \leq 25$ and every support degree, and 19,434 exact scalar remainder inequalities (57) for $2 \leq N \leq 80$, $0 \leq M < N$, $0 \leq J \leq 5$; computes the symbolic operator expansion for general k through order four; verifies every coefficient of Part I through order three; and asserts the fourth-order values $\mathcal{P}_4(-1)$ and $q_4(-1)$. Validation uses explicit exceptions, so Python's `-O` does not disable it. With `--inject-error` it adds a spurious v^5 term to q_3 and must exit nonzero, both normally and under `-O`. The independent check uses neither H_s nor the Euler operator: it starts from the exact identity $R(R^n)' = nR'R^n$, divides each coefficient by $(N)_M$, takes the logarithm directly in v through w^4 and v^{12} for symbolic k , and enumerates all rank words for six small (n, k) . In the manuscript's words, the mathematical proofs, not these finite tests, establish the uniform estimates and all-orders conclusions.

Not shipped: the manuscript `repair_note.tex` (merged here) and its 7-page PDF, the delivery README, the checksum manifest and its verifier, a second copy of the independent output (byte-identical), the PDF build script, and the directory `source/`, which holds byte copies of this report's `README.md`, `article.tex`, `code/derive_symbolic.py` and `code/verify.py` as they stood at `4b874cea0`. The README of this report says how to rebuild that directory and rerun the checks.

The delivery's audit finds no remaining mathematical gap within the stated fixed-multiplicity scope, reproduces the fourth-order coefficient independently, and records two issues resolved before delivery (an integer tail cut, a stale file name in a docstring). It is a mathematical and code review, not a machine-checked formalization, a literature or priority review, or a certified numerical inverse; it checked the source files against the manifest only, without retrieving the repository.

III.11 Provenance of Part III

[Added 5 October 2026, batch 101.]

Source pin. The manuscript reviews this report at

`4b874cea0012c51a6841ad9c58f6e20fa57c71da,`

the pin of Part II as well. Its manifest records the Git blob identifiers of `article.tex` (`afa43ce6...`), `README.md` (`c9472b1c...`), `code/derive_symbolic.py` and `code/verify.py` at that commit, retrieved on 2 October 2026. The last two are unchanged at the time of this write; the first two were changed by the batch-77P5 write and again by this one, so their identifiers no longer match.

Intake record. Bundle Report 98 of batch 101, archive `A330266_Uniform_Tail_Repair.zip`; arrival commit `60f54ea06`; placement commit `f7c612c72`, which staged eight delivered files with

the prefix `03-tail-` and removed the archive from `docs/incoming` (it survives in `60f54ea06`). The manuscript has an empty author line and empty author metadata, and cites no literature beyond this report.

A stale sentence of the manuscript. [5 October 2026.] The manuscript says that this report’s README “explicitly records the missing analytic justification”. That was true at `4b874cea0`; since `1a372a1ca` (2 October 2026) the README states that the tail estimate is proved in Part II. Likewise its opening sentence that the gap “is closed” by its inequality describes the second closure, not the first.

Where the merge had to choose.

- Statements and proofs that agree with Part II step by step (Lemma 2, Theorem 3, Theorem 4, the inverse) are cited through the map of Section III.2, not printed again; formulas Part I or Part II already displays (q_1-q_3 , $p_{n,k}$ to N^{-3} , the local law, the ratio) are cited. Of the manuscript’s 24 labels, six are kept with the prefix `bsw:ut:` (`eq:tail`, `eq:C`, `eq:p4`, `eq:q4minus`, `eq:k4`, `sec:sources`) and 18 are dropped with the cited material.
- Kept as marked second routes: the marked-forest proof of (20) and the $(1 + \lambda z)^k$ proof of Theorem II.3.1.
- Printed in full as new relative to Parts I–II: the tail bound, the unequal-rank identity and bound, the sharper constant, the degree bound of $\mathcal{P}_s$, the fourth-order terms, the smooth-carrier inverse, and the second further question.
- Added by the write and marked [write]: the explicit proof of the unequal-rank identity, Proposition III.4.3, Remarks III.4.4, III.5.2, III.7.3, III.7.4 and III.7.5, the last two columns of the table in Corollary III.7.2, H_4 in display (77) (from the shipped coefficient file), the proof of Remark III.8.1, the sketch in question 3, and the notes.
- Symbols were renamed as listed in Section III.1; no formula changed its content. The manuscript’s Section 8 (“Sources and reproducibility”) is rewritten to name the shipped files. Nothing of the manuscript’s mathematics, scope statements or disclaimers was removed.

Checks by the intake. The intake verified the delivery’s checksum manifest (18 files) and the four source blobs against the pin, and reran both checks on copies (SymPy 1.14.0, Python 3.14 on Windows): the main check exits 0 normally and under `-0` with identical output, regenerating its two output files equal to the delivered ones apart from line endings; `--inject-error` exits 1 in both modes; the independent check exits 0 with output equal to the recorded one apart from line endings. On 5 October 2026 the write reran the main check from the shipped files with `source/` rebuilt from `4b874cea0`, as the README describes (exit 0 in 23 s, outputs equal apart from line endings). Independently of the delivered programs: Richardson extrapolation of n^4 times the residual of $e^{k-1}p_{n,k}$ after the n^{-3} term of (42), from the exact counts at $n = 100, 200, 400$, gives 0.058755, 0.150888, 0.225673, 0.229544, 0.119915 for $k = 2, \dots, 6$, against the claimed 0.058757, 0.150892, 0.225677, 0.229547, 0.119917; the residuals recorded in Part I’s README (0.228 and 0.226 at $n = 100, 400$ for $k = 4$) and the third-order residuals of the table in Section 9.2 (within 4% of $\mathcal{P}_4(-1)/(k^4 n^4)$ at $n = 50$ for every k) agree. The unequal-rank identity and bound were confirmed by brute force for the compositions $(1, 2, 3)$, $(2, 3)$, $(1, 1, 4)$, $(3, 3, 1)$, $(2, 2, 3)$, $(1, 4, 2)$, $(5, 2)$; the factorial moments of the composition $(1, 3, 2, 3)$ repeated $n/4$ times, at $n = 20, 80, 320$, approach μ_n^M for $M \leq 4$ at the rate $O(n^{-1})$; and the manuscript’s ratio coefficients equal those of (52) for $k = 2, \dots, 7$. These checks do not replace an independent review of the proofs.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of Part III, made by the intake after the write (a75a4e41b), reread every item printed in full and every addition of the write: Theorem III.7.1 and the note after it, Corollary III.7.2 with the logarithmic line for $k = 4$, the Lah-number note of Section III.3 (with its pointer to Section 11 of Part I), Propositions III.3.1, III.4.2 and III.4.3, Corollary III.4.1, Remarks III.5.1, III.5.2, III.6.1, III.6.2, III.7.3, III.7.4, III.7.5 and III.8.1, and the batch-101 notes in Section 9 of Part I. It found every item valid, no counterexample, and no gap in any proof chain. Proposition III.3.1 was valid with three defects of wording (the case $M > N$, the derivation of the mean when every $k_i = 1$, and the implicit $n \geq 1$); they are corrected where they stand, with a dated note keeping the first wording. In the proof of Proposition III.4.3 the bound “each product is at most $(2^K K^M)^M$ ” was true but loose; it now reads $2^{KM} K^M$, the sharp form of the same argument, and that argument is complete: a factor with $j_i = 0$ equals $a_{k_i,0} = 1$, each of the at most M other factors is at most $2^K K^{j_i}$, and the j_i sum to M . The statement of that proposition is unchanged.

The check used neither the delivered programs nor the write’s. Its own symbolic derivation (SymPy 1.14, symbolic k) built $H_r(M)$ from $\exp \sum_j x^j \sigma_j(M)/j$ by Faulhaber’s formula, $b_1, \dots, b_5$ from $\log R_k$, the f_s from $F(w, v)$, the $\mathcal{P}_s$ by its own implementation of $e^{-\lambda v} H_r(D) e^{\lambda v} = H_r(D + \lambda v)$, and the q_s as the coefficients of $\log \sum_s \mathcal{P}_s w^s$. It reproduces exactly (77) (also by brute force for $M \leq 11$), q_1, q_2, q_3 of (70), (78), (79), the displayed $\mathcal{P}_4(-1)$, (81), $\mathcal{P}_1(-1), \mathcal{P}_2(-1), \mathcal{P}_3(-1)$ of (42), $b_1, \dots, b_4$ of Part II together with $b_5 = 2k\lambda^2(7k - 10)(3k^2 - 8k + 6)/5$, every entry of the table in Corollary III.7.2, and $A_1, \dots, A_4$. The leading coefficient of $\mathcal{P}_s$ is $(-1)^s \lambda^{2s}/(2^s s!)$ for $s \leq 4$, so $\deg \mathcal{P}_s = 2s$ there; $\deg q_4 = 5$ for $3 \leq k \leq 11$ and $\deg q_4 = 4$ for $k = 2$; and $q_1, \dots, q_4$ have zero constant and linear coefficients.

The numerical tests use no theory. The check computed $r_{n,M} = [z^M] R_k(z)^n$ exactly in big integers, and $B_{n,k}$ exactly by grouped inclusion–exclusion, checked against enumeration of all rank words ($k = 2, n \leq 4$; $k = 3, n \leq 3$) and against $B_{13,4}$ of A330266. Least-degree (Richardson) fits in $1/N$ over the 32 consecutive values $n = 160, \dots, 191$, with mpmath at 260 digits, of $e^{-\lambda v} \Phi_{n,k}(v)$ and of its logarithm at the nine points $v \in \{-1, -\frac{3}{4}, -\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$ for $k = 2, \dots, 6$ recover $\mathcal{P}_s(v)$ and $q_s(v)$, $s = 1, \dots, 4$, to $1.5 \cdot 10^{-37}$ or better (typically 10^{-45} to 10^{-70}), and at $v = -1$ for $k = 7, \dots, 10$ to $4 \cdot 10^{-32}$. Nine points fix a polynomial of degree 8, so the tests reach every coefficient of $\mathcal{P}_4$ and q_4 as polynomials in v for each $k \leq 6$, and, with $k = 2, \dots, 10$, every coefficient of $\mathcal{P}_4(-1)$ (degree 8 in k) and of $q_4(-1)$ (degree 5). Because these fits start from exact counts, they test the operator method as well as the algebra.

For unequal ranks, enumeration of all rank words of the compositions $(1), (1, 1, 1), (2, 1), (1, 2, 3), (2, 2, 1, 1), (3, 1), (4, 2), (3, 3, 2), (2, 2, 2, 2), (3, 3, 3), (1, 1, 4)$ and $(5, 1, 1)$ confirms (73) for $M \leq N - n$, $\mathbb{E}(\frac{X_{\mathbf{k}}}{M}) = r_{\mathbf{k},M} = 0$ for $N - n < M \leq N + 1$, the maximum $N - n$ (attained), the mean, and (75). For Proposition III.4.3: for the ranks $(1, \dots, 1, 3)$, $\mathbb{E}(X)_{\underline{3}} = 0$ while $\mu_n^3 = 216/N^3$, as the case $M > N - n$ of the proof requires; and for three random compositions per n with $k_i \leq K$, $n = 25, \dots, 400$ and $M \leq 5$, the quantity $n|\mathbb{E}(X_{\mathbf{k}})_{\underline{M}} - \mu_n^M|$ stays between 1.8 and 2.2 for $K = 2, 33$ and 43 for $K = 3$, and 118 and 258 for $K = 4$. For Remark III.5.2, over 40 random compositions with $k_i \leq 5$ and $n \leq 12$, $J = 0, \dots, 3$, $V \in \{\frac{1}{2}, 1, 2, 4\}$ and 64 points on $|v| = V$, the left side is at most 0.43 times the right. For Remark III.7.3, through the carrier of Remark III.8.1, $(N_*(B_{n,k}) - N)N^4 \log N$ at $n = 80$ is 0.773 for $k = 2$ and 1.93 for $k = 4$ (still decreasing, slowed by a large A_5), against $A_4 = \frac{3}{4}$ in both cases. For Remark III.7.4, the exact $p_{13,4}$, the two approximations and the four errors reproduce as printed; in the table of Section 9.2 of Part I, the third-order residuals at $n = 50$ are 1.036, 1.031, 1.024, 1.018, 1.012 times $\mathcal{P}_4(-1)/(kn)^4$ for $k = 2, \dots, 6$, and the table’s values at $n = 10, 50$ reproduce to every printed digit. For Remark III.7.5 with $J = 4$, against the exact $\mathbb{P}(X = j)$,

N^5 times the error is stable at $n = 50, 100, 200$ for $j = 0, \dots, 3$ and $k = 2, 4$ (for example 1.257, 1.241, 1.234 for $k = 2, j = 0$), and the case $J = 1$ reproduces (43) by hand. For Remark III.8.1, at $Y = B_{n,k}$ with $n = 5, \dots, 80$, the error of the displayed approximation to N_* , times $N_0 \log N_0$, stays between 0.10 and 0.11 for $k = 2$ and falls from 0.31 to 0.12 for $k = 4$.

Report 98's manuscript `repair_note.tex` is not shipped (Section III.10), and the check did not see it. Where Part III attributes a statement or a claim to the manuscript, for instance the claim on further local corrections that Remark III.7.5 proves, the check took the attribution from Part III's own description. This was a careful reading with numerical tests, not a formal verification or an external review.

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